

Pega Platform 26.1.1 on Azure Kubernetes Service

Externalized Kafka (Confluent Cloud) and Search (SRS with Elasticsearch or OpenSearch)

Design, deployment, configuration and migration from Pega Platform 8.8, including Kafka topic and search index migration requirements, validation, strategy and execution

Document version	1.0
Status	Issued for customer review and approval
Issue date	1 October 2026
Classification	Customer Confidential
Target release	Pega Platform 26.1.1 (Pega Platform '26), client-managed cloud on Azure AKS
Source release	Pega Platform 8.8

Document control

Item	Detail
Title	Pega Platform 26.1.1 on Azure AKS: externalized Kafka and search. Design, deployment, configuration and migration runbook
Version	1.0
Status	Issued for customer review and approval
Issue date	1 October 2026
Evidence cut-off	Vendor documentation was read on 1 October 2026. Section 3 lists the version facts used and Appendix F lists every source.
Supersedes	The draft integrated customer engineering runbook (sections 1 to 30, decisions D-01 to D-12, open decisions OD-01 to OD-15). This document replaces its Kafka, search and upgrade content and rennumbers the open decisions (Section 22).
Owner	Customer platform engineering lead
Review cycle	Before each rehearsal and before each production release (Section 13)

Revision history

Version	Date	Change	Author
0.1 to 0.9	2026	Draft integrated runbook (customer engineering).	Customer engineering
1.0	1 October 2026	Rewritten and evidence-checked runbook. Adds the 8.8 upgrade path, Kafka topic and search index migration, decision diagrams, validation matrices and appendices.	Implementation team

Approvals

Approval confirms that the approver has reviewed the sections relevant to the role and accepts the recommendations and open decisions as stated.

Role	Name	Signature	Date
Customer business owner			
Customer enterprise architect			
Customer security architect			
Customer platform engineering lead			
Pega Lead System Architect			
Kafka and search engineering lead			

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1 Executive summary

1.1 Purpose

This runbook explains how to build Pega Platform 26.1.1 on Azure Kubernetes Service (AKS) with Kafka provided by Confluent Cloud and search provided by the Pega Search and Reporting Service (SRS) backed by an Elasticsearch or OpenSearch cluster. It also explains how to move the current Pega Platform 8.8 system to that platform, with particular attention to two questions the customer asked: what happens to Kafka topics, and what happens to search indexes. For each, the runbook states the requirement, the validation needed, the strategy chosen and the steps to execute it.

1.2 Facts that shape the plan

The plan follows from six published facts. Each one is taken from vendor documentation and referenced.

Table 1. Published facts that shape the plan

#	Fact	Consequence for this programme	Source
1	From Pega Platform '24.2, embedded Kafka, Elasticsearch and Cassandra are not supported. Kafka and search must be external.	Pega 26.1.1 needs Confluent Cloud and SRS in place before it can run.	[R2]
2	From Pega Platform '25, only containerized deployments are supported, on Tomcat 10.1 (Jakarta EE).	Custom JARs and CodeSets that use Java EE (javax) packages must be rebuilt for Jakarta EE.	[R2, R3]
3	Hazelcast is removed in Pega Platform '25. It must be removed before updating to '25 or later. The removal tooling exists only in patches 23.1.4, 23.1.5, 24.1.3, 24.1.4, 24.2.2, 24.2.3 and later.	Pega 8.8 cannot be updated directly to 26.1.1. An intermediate (bridge) release is needed.	[R5, R9]
4	All tables must have primary keys before an update to '25 or later.	Run the primaryKeyUtility on 8.8 in dry-run mode early and fix gaps.	[R3, R12]
5	Stream data stored in Kafka cannot be moved to a new Kafka provider. Pega provides a Stream Migration activity (8.7 and later) that drains queues before the switch.	Kafka topics are not copied. They are drained on the source and created empty on Confluent.	[R15]
6	After Pega connects to SRS, SRS indexes all searchable data from the database. This needs a downtime period.	Search indexes are rebuilt from the database, not copied. The build time is measured in rehearsal.	[R17]

1.3 Recommended approach

The recommended path has three production releases. Each release changes one class of thing, so a failure has one likely cause and a simple rollback.

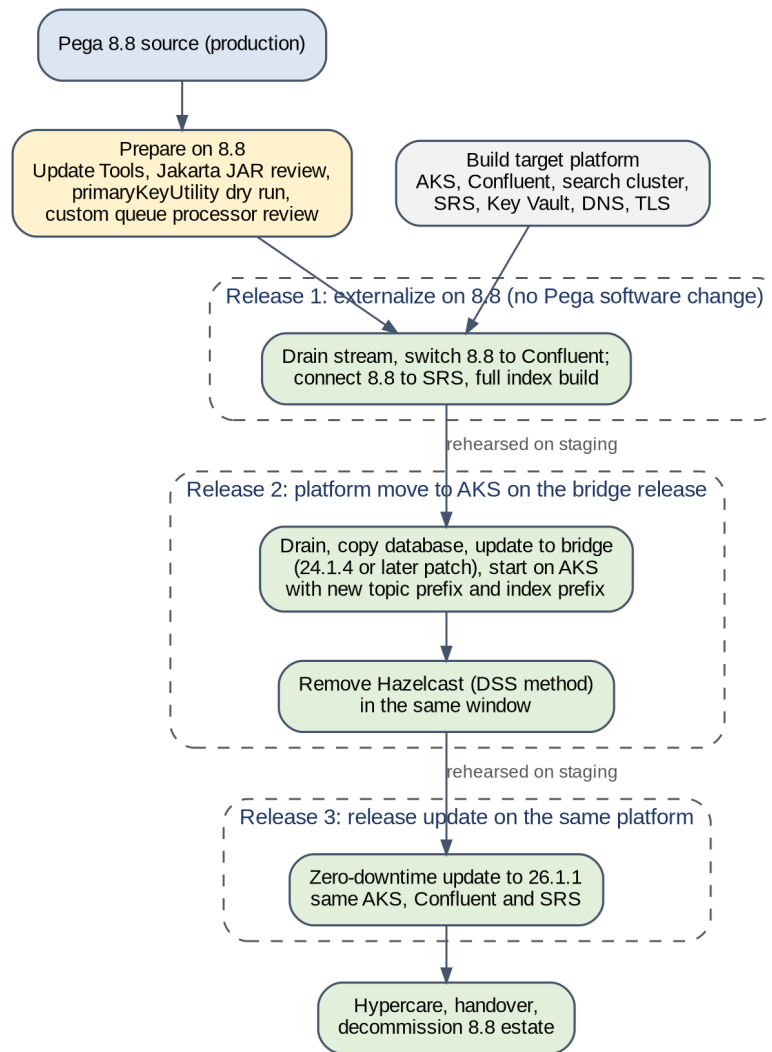


Figure 1. Recommended three-release path from Pega 8.8 to Pega 26.1.1

Prepared for this implementation. Not a Pega-published diagram.

Table 2. The three releases

Release	What changes	What does not change	Main evidence of success
Release 1: externalize on 8.8	8.8 switches from its current stream service to Confluent Cloud, and from its current search to SRS. Full index build.	Pega software version, servers, database.	Stream landing page shows Provider ExternalKafka and Status NORMAL [R15]. Index status complete for all classes [R21].
Release 2: platform move	Database copied to Azure and updated to the bridge release (latest '24.1 patch, 24.1.4 or later). Pega runs on AKS. Hazelcast removed.	Confluent cluster and search cluster (new topic prefix and new index prefix on them).	Hazelcast decommission APIs report success [R8]. Smoke and regression tests pass.
Release 3: release update	Zero-downtime update from the bridge release to 26.1.1.	AKS cluster, Confluent topics, SRS indexes.	Installer job completes. Stream and search checks pass on 26.1.1.

Pega's own published sequence for clients on older releases is the same in substance: update to the latest '24.1 patch, externalize Kafka, search and (where used) Cassandra, remove Hazelcast, move to containers, then update to '25 or later [R2]. Release 1 moves externalization ahead of the software update because Pega

advises migrating from embedded stream to external Kafka before a software update, and not changing the Pega version during a stream provider switch [\[R30, R33\]](#).

1.4 Decisions the customer needs to take

The following open decisions affect cost, timeline or security and need an owner and a date. Section 22 lists all of them.

Table 3. Open decisions that need an early answer

ID	Decision	Recommended answer
OD-02	Bridge release patch	Latest '24.1 patch (24.1.4 or later), confirmed with Pega Support.
OD-03	Three releases or a combined Release 1 and 2	Three releases.
OD-04	Search provider	Managed OpenSearch from an Azure-hosted provider, or Elastic Cloud on Azure if Elastic licensing is accepted. Both meet the SRS matrix [R16] .
OD-05	Confluent Cloud cluster type	Enterprise with Private Link, or Dedicated if the partition budget or networking needs exceed Enterprise limits [R37] .
OD-06	Kafka authentication	SASL_SSL with PLAIN (Confluent API key in Key Vault) as the documented Helm path [R28] .
OD-16	Identity provider for Pega-to-SRS OAuth	Choose a provider that can issue the scope and guid claims SRS needs; prove it in the first environment.

1.5 Readiness statement

This document is ready for customer review and approval as a design and runbook. It does not replace the rehearsals it prescribes. Durations for draining, database copy, update and index build depend on data volumes and must come from the staging rehearsal (Section 16), not from estimates. Where vendor sources disagree, the conflict is recorded in Section 23 with the action taken.

2 Scope and how to use this document

2.1 Scope

Table 4. Scope of this document

In scope	Out of scope
Pega Platform 26.1.1 on AKS using the Pega Helm charts (pega and backingservices). Confluent Cloud as the Pega stream service: cluster type, networking, security, topics and Helm configuration. SRS with Elasticsearch or OpenSearch: provider choice, sizing, security and Helm configuration. Upgrade path from Pega 8.8, including Hazelcast removal, Tomcat 10.1 and primary key prerequisites. Kafka topic migration and search index migration: requirements, validation, strategy and execution. Cutover, rollback, testing, troubleshooting, observability and operations for these components.	Pega application design changes and functional regression content (owned by the application team). Cassandra and Decision Data Store. Cassandra is required only for Pega Customer Decision Hub and Pega Process AI [R22]. If either is licensed, a separate plan is needed. Constellation UI service and Pega Diagnostic Center onboarding beyond connectivity. Database engine migration. The database engine and version must be on the 26.1.1 Platform Support Guide [R1]. Commercial terms with Confluent, Elastic or OpenSearch providers.

2.2 Audience and reading guide

Table 5. Reading guide by role

Reader	Read first	Then
Business owner, programme manager	Section 1, Section 10.3, Section 14, Section 21, Section 22	Section 15
Enterprise and security architects	Sections 4, 5, 7.5, 8.8, 9	Sections 21 to 23
Pega Lead System Architect	Sections 10, 11, 12	Sections 14 to 17
Platform, Kafka and search engineers	Sections 6 to 9 and 13	Sections 17, 18, Appendices B and D
Test lead	Section 16	Validation matrices in Sections 11.5 and 12.5

2.3 Conventions

- Must** marks a vendor requirement or a fixed customer decision. **Should** marks a recommendation that can be changed through the decision process in Section 5.
- Values in angle brackets, for example `<env>` or `<bootstrap-host>`, are environment-specific. Each one is listed in the configuration inventory in Appendix A and recorded before deployment.
- References in square brackets, for example [R13], point to the numbered sources in Appendix F.
- "Bridge release" means the intermediate Pega release between 8.8 and 26.1.1. "Release 1", "Release 2" and "Release 3" mean the three production changes in Figure 1.
- "Search cluster" means the Elasticsearch or OpenSearch cluster that stores SRS indexes. "Stream" means the Pega stream service, which uses Kafka.
- Figures marked "Prepared for this implementation" were drawn for this document. Figures from Pega, Pega Academy, Confluent and OpenSearch are reproduced with attribution (Appendix G).

2.4 How the evidence was gathered

Every statement about product behavior comes from one of these sources, in this order of authority:

1. Pega documentation for Pega Platform '26 (docs.pega.com), and for '25 or '24.1 where the topic belongs to that release (for example, Hazelcast removal).
2. The Pega Helm charts repository on GitHub (chart version 4.13.0) for configuration keys and defaults.
3. Pega Academy for architecture illustrations and deployment patterns.
4. Confluent, OpenSearch, Microsoft and Apache Kafka documentation for their own products.

Where two sources disagree, the higher one is followed and the conflict is logged in Section 23. Where no source settles a point, the document says so and turns it into a validation step or an open decision.

3 Evidence baseline: versions and support facts

The table records the facts this design depends on, as published on the evidence cut-off date. Re-check them before each release, because vendors update these pages.

Table 6. Version and support facts used in this design

Topic	Fact	Source
Target release	Pega Platform '26; the Platform Support Guide covers 26.1.x.	[R1]
Kubernetes	AKS is a supported environment. x86-64 nodes only (no ARM). At least two worker nodes with 32 GB RAM. Database in the same region as the worker nodes. Tomcat must be the one in the Pega image.	[R22]
Helm	Helm 3.0 or later. Pega Helm charts 4.13.0 used for this design.	[R22, R28]
Kafka requirement	"All Queue Processors and Job Schedulers in Pega Platform require Kafka." Without it, Pega Platform is not fully functional.	[R13]
Kafka client	Pega '26 ships the Kafka client library 4.0.0, backward compatible with 3.9.2, and supports Kafka server 4.0.0.	[R3, R22]
Kafka settings	message.max.bytes 5000000; replica.fetch.max.bytes 5000100; replica.fetch.response.max.bytes 5000100; unclean.leader.election.enable false; auto.create.topics.enable false.	[R13]
Kafka security	Generic Kafka settings only; vendor-specific features such as IAM roles are not supported. Mechanisms: SASL using JAAS, SASL/OAUTHBEARER, SASL/PLAIN, SASL/SCRAM.	[R13]
Kafka ACLs	TOPIC and GROUP on the streamNamePattern prefix (ALL, PREFIXED); TRANSACTIONAL_ID * (READ/WRITE, LITERAL); CLUSTER IDEMPOTENT_WRITE (LITERAL).	[R13]
Confluent version	If Confluent is the provider, use 5.4.x or later, compatible with the supported Kafka version.	[R13]
Stream migration	Stream Migration activity available from 8.7. Drains RUNNING queue processors and data flows only. Does not transfer stored data.	[R15]
Search	SRS replaces embedded search and the legacy plug-in (deprecated in 8.8, removed in '24.2).	[R16]
SRS matrix	Pega 8.6 and later: SRS 1.44.3 or later. Elasticsearch with authentication: 7.17.9, 7.17.29, 8.10.3, 8.15.1, 8.15.5, 8.18.2, 8.18.3, 8.19.11. OpenSearch image: AWS Elasticsearch 7.10, OpenSearch 1.3, 2.15, 2.19.	[R16]
Search settings	Index auto-creation off (SRS sets it if its user has the manage cluster privilege); action.destructive_requires_name false.	[R16]
Index build	After connecting to SRS, SRS indexes all searchable data; this requires downtime.	[R17]
Hazelcast	Removed in '25. Remove before updating. Removal needs 23.1.4, 24.1.3, 24.2.2 or later patches.	[R5, R9]
Hazelcast prerequisites	About 100 additional Kafka partitions per environment; database CPU buffer of at least 10 %; Kafka CPU buffer of at least 15 % below alert thresholds.	[R6]
Tomcat	Tomcat 10.1 only from '25. Rules referencing Java EE convert automatically; custom CodeSets and JARs must be updated.	[R3, R11]
Primary keys	All tables need primary keys before updating to '25 or later (primaryKeyUtility).	[R3, R12]
Queue processing	'26 moves to partitioned, multi-threaded processing. Remove DSS delayeditems/dataflowbased/threadspernode (PEGA0179).	[R3]
System Pulse	Set SystemPulse_SystemPulseTopicName to 6 partitions if dynamic topic creation is restricted.	[R3]
Zero-downtime update	Helm upgradeType zero-downtime, for updates from 8.4.2 and later.	[R28, R33]

4 Current state and target architecture

4.1 Source estate assessment

Several choices in this runbook depend on facts about the 8.8 estate. Collect them first and record the answers in Appendix A. Each row names the check and why it matters.

Table 7. Source estate assessment checklist

Item	How to check	Why it matters
Exact 8.8 patch	Dev Studio: About Pega Platform; or the installer log.	Sets the starting point of the update path and any hotfix prerequisites.
Deployment type	Infrastructure inventory: VMs or Kubernetes.	VMs are not supported from '25 [R2] ; affects how Release 1 is configured.
Stream mode	Configure > Decisioning > Infrastructure > Services > Stream. Provider shows embedded stream nodes or ExternalKafka.	Decides how Release 1 drains and switches the stream (Section 11.4).
Search mode	Search landing page and prconfig or Helm values: embedded, legacy plug-in, or SRS.	Embedded and plug-in data cannot be reused; a full SRS index build is needed (Section 12).
Hazelcast mode	Helm values (hazelcast.enabled, hazelcast.clusteringServiceEnabled) or prconfig.	Embedded Hazelcast or the Clustering Service changes the removal steps [R8] .
Database engine and version	DBA inventory.	Must be on the 26.1.1 Platform Support Guide [R1] .
Custom JARs and CodeSets	Pega Update Tools and the Tomcat 10.1 checks [R11] .	Java EE packages must move to Jakarta EE for '25 and later.
Tables without primary keys	primaryKeyUtility in dry-run mode [R12] .	Blocks the update to '25 and later.
Custom queue processors and DSS	Records > SysAdmin > Dynamic System Settings; queue processor rules.	PEGA0179 change in '26; DSS delayeditems/dataflowbased/threadspnode must be removed [R3] .
Kafka data sets	Records > Data Model > Data Set, type Kafka; Kafka configuration instances.	Application Kafka integrations are separate from the stream service and are handled in Section 11.4.
Searchable data volume	Search landing page per class; database row counts of indexed classes.	Drives search cluster sizing (Section 8.4) and index build time.
Queue backlog	Admin Studio > Queue processors: Ready to process, Broken.	Large backlogs lengthen the drain step; broken items must be resolved or accepted before the drain.
Licensed Pega applications	Application stack in Dev Studio.	Customer Decision Hub or Process AI bring Cassandra and Kafka data set needs [R13, R22] .

4.2 Pega reference architecture for '26

Pega's published Kubernetes architecture for '26 has three layers: the Pega application namespace (web and batch deployments plus an installer job), backing services that Pega provides (SRS and Constellation), and technologies the client provides (Elasticsearch or OpenSearch, Kafka, Cassandra for CDH only, and the database) [\[R23\]](#). Hazelcast does not appear, because '25 removed it.

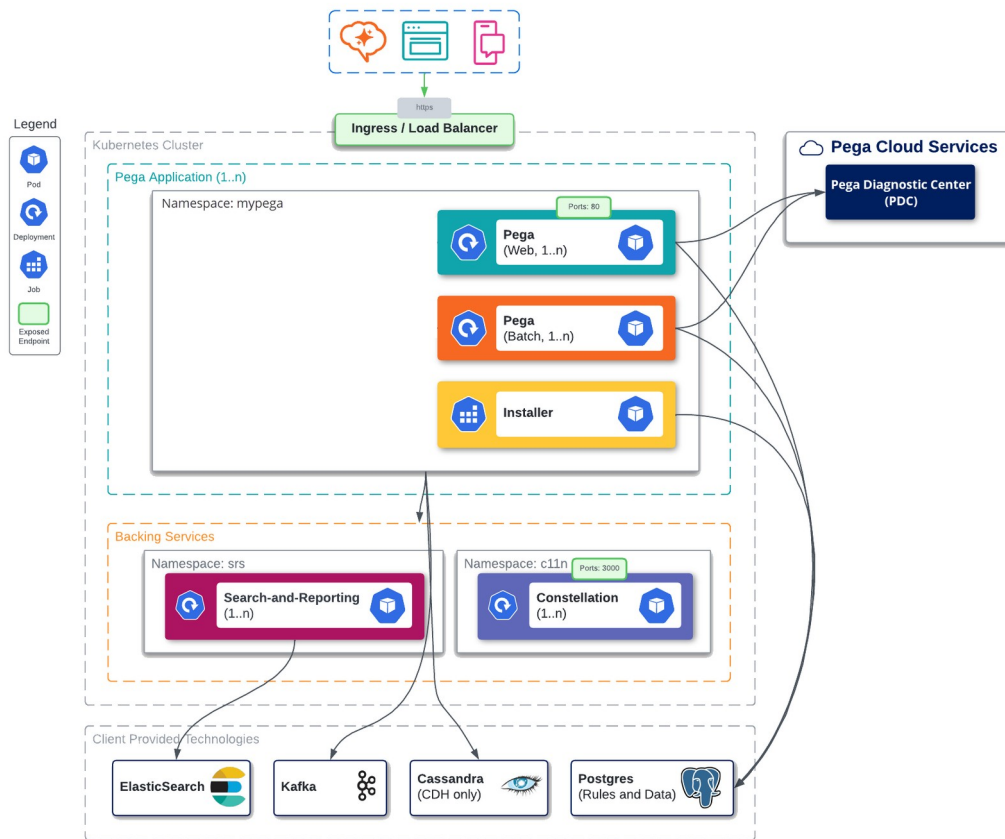


Figure 2. Pega Platform Kubernetes architecture for '26

Source: Pega Documentation, "Pega Platform Kubernetes architecture" [R23]. © Pegasystems Inc. Reproduced with attribution. The database shown is an example; any database on the Platform Support Guide can be used.

Pega Academy describes the same move from embedded to external services. Before externalization, Kafka, Cassandra and Elasticsearch ran inside the Pega cluster. After it, they are separate services the client operates, and Pega reaches search only through SRS [R35].

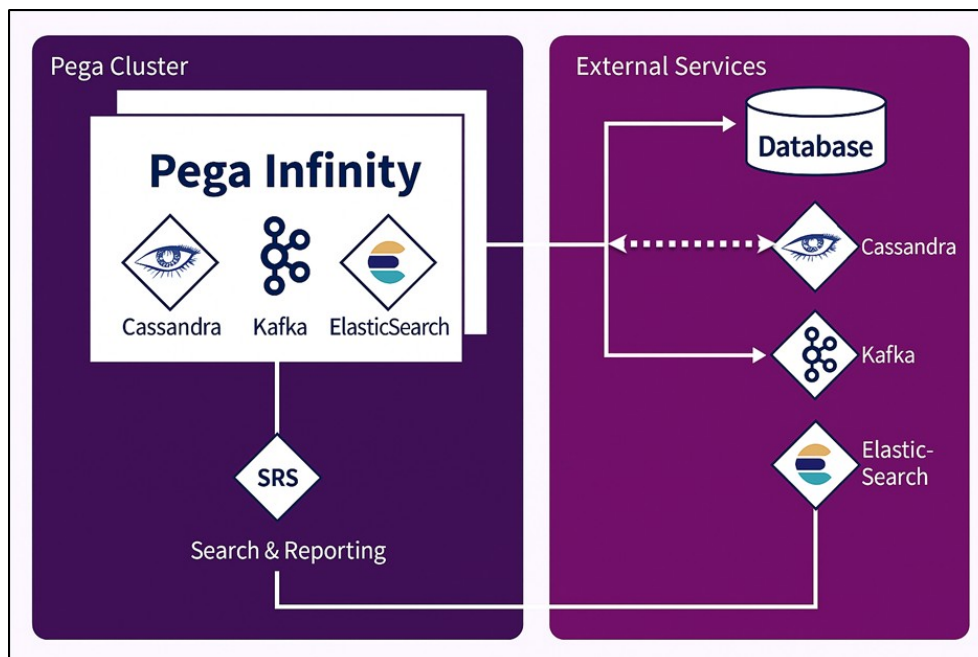


Figure 3. Externalized services in a Pega deployment

Source: Pega Academy, "Deployment architecture with external services" [R35]. © Pegasystems Inc. Reproduced with attribution.

4.3 Target architecture on Azure

Figure 4 shows the target. Pega web and batch tiers, the installer job and SRS run in a private AKS cluster in the customer subscription. Every managed service is reached through a private endpoint in a dedicated subnet. Users reach Pega through Azure Application Gateway with Web Application Firewall. Secrets come from Azure Key Vault through the External Secrets Operator. Outbound traffic leaves through the hub firewall.

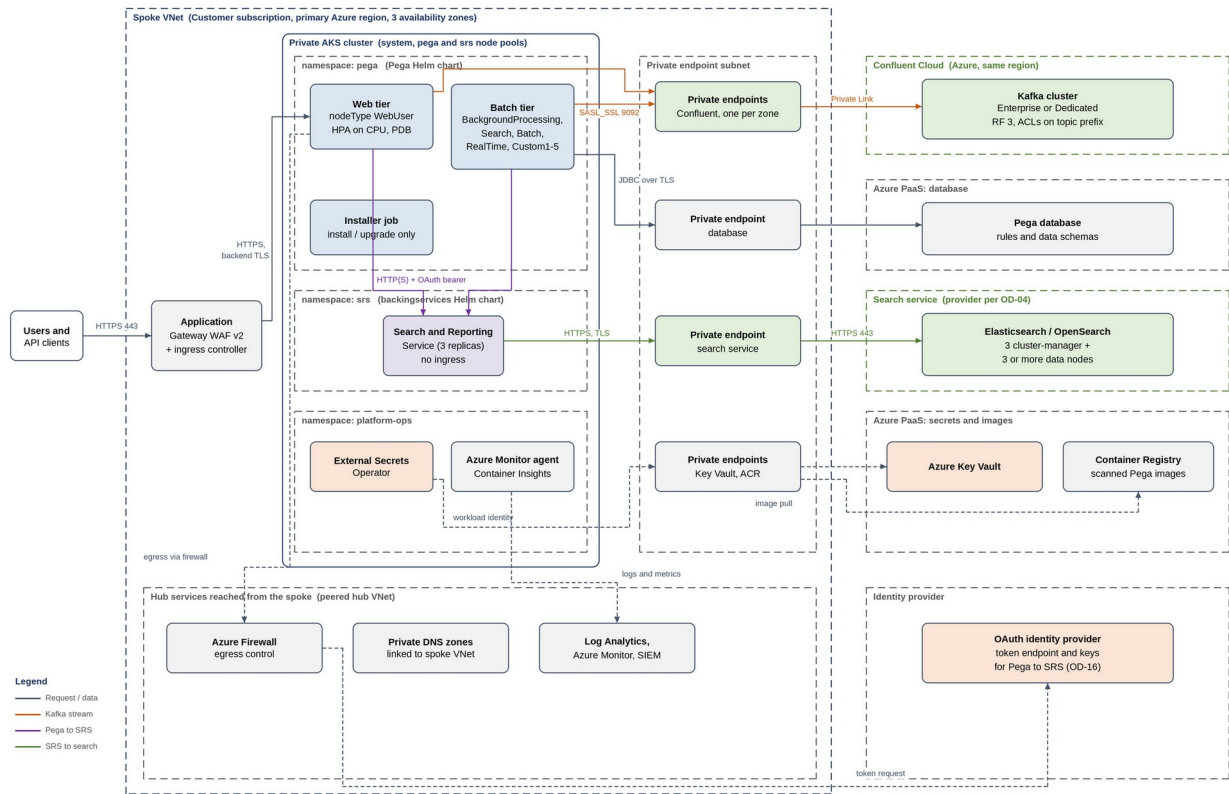


Figure 4. Target architecture on Azure AKS
Prepared for this implementation. Not a Pega-published diagram.

Table 8. Target architecture components

Component	Role	Design choice	Reference
Application Gateway WAF v2 and ingress controller	Public entry point; TLS termination and re-encryption to the web tier	Backend TLS to the web pods (tier.service.tls); request timeout set with appgw.ingress.kubernetes.io/request-timeout	[R28]
Web tier	User and API traffic (nodeType WebUser)	HPA on CPU; PDB enabled	[R28]
Batch tier	Background processing, search indexing, batch, real-time and custom node types	No ingress; sized for queue processor load	[R23, R28]
Installer job	Install and update the database schemas	Runs only during install or update	[R28]
SRS	Search and Reporting Service between Pega and the search cluster	Three replicas, no ingress, network policy enabled, OAuth enabled	[R16, R29]
External Secrets Operator	Synchronizes Key Vault secrets into Kubernetes secrets	Workload identity to Key Vault	[R28, R47]
Confluent Cloud	Kafka for the Pega stream service	Enterprise or Dedicated, private connectivity, RF 3	[R13, R37, R38]

Component	Role	Design choice	Reference
Search cluster	Elasticsearch or OpenSearch storage for SRS	Three cluster-manager and three or more data nodes; private connectivity	[R16]
Pega database	Rules and data schemas	Same region as AKS; private endpoint; TLS	[R22]
Key Vault, Container Registry	Secrets and certificates; Pega images	Private endpoints; images pulled from Pega and scanned	[R28]

4.4 Logical data flows

Figure 5 shows the five flows that matter for Kafka and search. Flow 1 is the user request. Flow 2 is queue processing: producers on any tier write to Kafka topics and the batch tier consumes them. Flow 3 is search: the batch tier indexes changed records through SRS, and the web tier sends queries through SRS. Flows 4 and 5 are administration and telemetry.

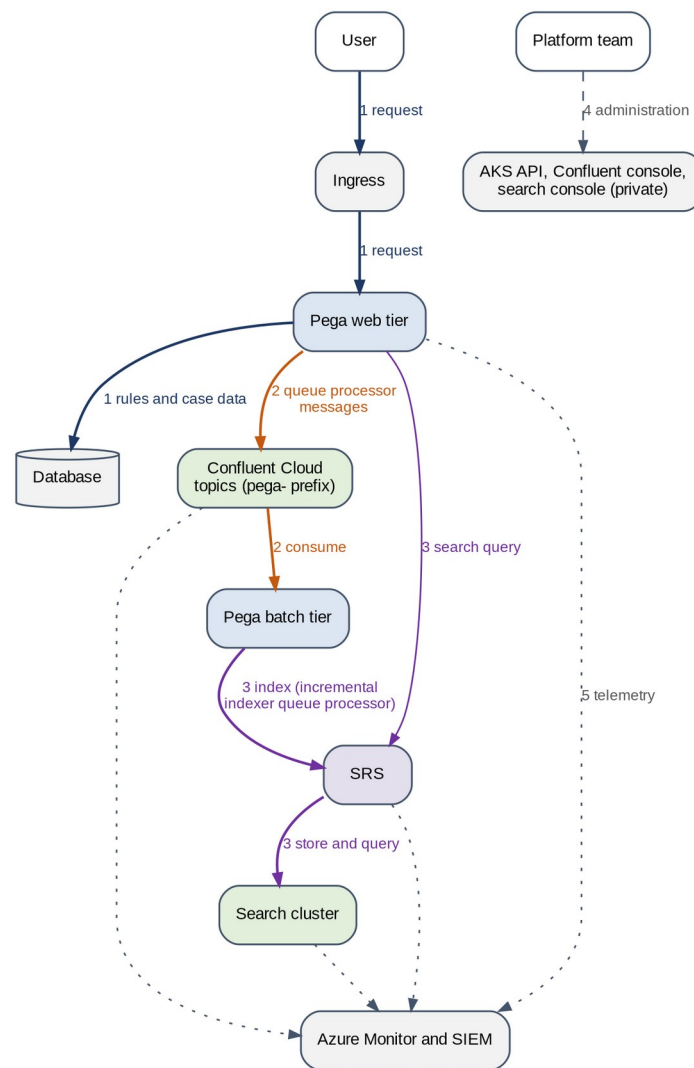


Figure 5. Logical flows between Pega, Kafka, SRS and the search cluster
Prepared for this implementation. Not a Pega-published diagram.

4.5 Tiers, namespaces and node pools

Table 9. Namespaces, charts and node pools

Namespace	Workload	Chart	Node pool	Notes
pega - <env>	Web tier, batch tier, installer job	pega	pega (user pool, 3 zones)	Pega namespaces in Pega's architecture are named mypega in examples [R23] .
srs - <env>	SRS deployment and service	backingservices	srs (user pool) or shared user pool	SRS can serve more than one Pega environment through customerDeploymentId isolation [R16] .
platform-ops	External Secrets Operator, monitoring agents	vendor charts	system or user pool	Kept separate from Pega to limit RBAC scope.
kube-system	AKS system components	AKS managed	system pool	No Pega workloads.

5 Decision framework

5.1 Fixed customer decisions carried forward

The draft runbook recorded twelve fixed decisions (D-01 to D-12). Three of them directly constrain the Kafka and search design and are applied throughout this document.

Table 10. Fixed customer decisions that constrain Kafka and search

ID	Decision	Effect in this document
D-07	Azure Event Hubs is not used as the Pega Kafka service.	Confluent Cloud is the Kafka provider. This also matches Pega's statement that vendor-specific security features are not supported [R13] .
D-08	No Azure-managed Elasticsearch service.	The search provider is chosen from the options in Section 8.3 (OD-04).
D-09	No plain-text secrets in Helm values.	All passwords, JAAS strings and keys are delivered through <code>external_secret_name</code> and the External Secrets Operator (Section 9).

5.2 Architecture decisions made in this document

Table 11. Architecture decisions

ID	Decision	Choice	Reason	Source
AD-01	Upgrade path	Three releases with a bridge release	Hazelcast removal tooling is not available on 8.8; stream provider switch should not be combined with a software update	[R5, R30, R33]
AD-02	Bridge release	Latest '24.1 patch (24.1.4 or later)	Pega's documented adoption sequence names '24.1; 24.1.4 includes the Hazelcast removal changes without a hotfix	[R2, R7]
AD-03	Kafka topics	Drain and recreate; never copy Pega stream topics	Pega does not support moving stream data between providers	[R15, R30]
AD-04	Search indexes	Rebuild through SRS from the database	SRS indexes all searchable data after connection; embedded and plug-in data are not reusable	[R17]
AD-05	Topic prefix at platform move	New streamNamePattern prefix in Release 2	Keeps the 8.8 topics untouched for rollback	[R28]
AD-06	Index prefix at platform move	New customerDeploymentId in Release 2	Keeps the 8.8 indexes untouched for rollback; SRS isolates data by this ID	[R16, R28]
AD-07	Confluent networking	Private Link from the customer VNet	Private connectivity for production data	[R38]
AD-08	Kafka authentication	SASL_SSL with PLAIN (service account API key)	The documented Helm <code>saslMechanism</code> values are PLAIN, SCRAM-SHA-256 and SCRAM-SHA-512	[R28]
AD-09	Secrets	Key Vault with External Secrets Operator	Supported by the Pega Helm chart; satisfies D-09	[R28]
AD-10	Hazelcast removal method	Downtime method (DSS) inside the Release 2 window	Release 2 already has an outage; embedded Hazelcast needs no Clustering Service teardown	[R8]
AD-11	Release 3 update type	zero-downtime	Supported from 8.4.2 and later; same platform and services	[R28, R33]
AD-12	Search cluster sharing	Pega environments only; no non-Pega workloads	Pega best practice to avoid noisy neighbors	[R16]

5.3 Delivery option decision

Figure 6 shows how the delivery option is chosen. The first question is settled for 8.8: it is not on a patch that can remove Hazelcast. The second question checks whether 8.8 can reach Confluent and SRS privately, which Release 1 needs. If it cannot, or if the business will not accept three windows, Releases 1 and 2 can be combined. That combined option changes the stream provider and the Pega version in one window, which goes against the Helm chart guidance [R30]; it should be used only after Pega Support confirms it for this estate.

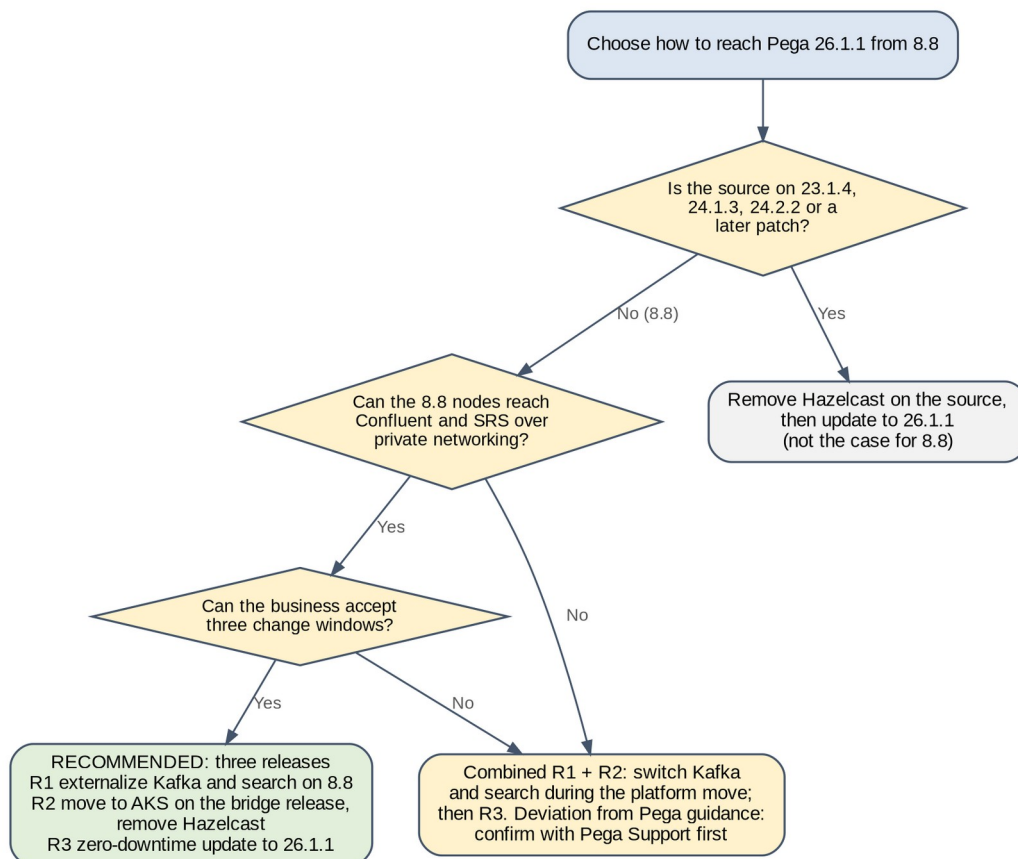


Figure 6. Decision: delivery option from 8.8 to 26.1.1

Prepared for this implementation. Not a Pega-published diagram.

5.4 How open decisions are handled

Each open decision in Section 22 has an owner, a due point in the plan and a default. If a decision is not taken by its due point, the default applies and the risk is recorded in Section 21. A decision that changes a fixed decision (D-xx) needs approval from the customer enterprise architect and the security architect.

6 Azure AKS platform design

The Pega Helm charts repository publishes a guide to deploying Pega Platform on AKS [R32]. This section applies that guide and the Pega client-managed cloud requirements [R22] to the customer's Azure landing zone.

6.1 Pega requirements and how the design meets them

Table 12. Pega platform requirements and the AKS design

Pega requirement	Design	Source
x86-64 CPUs only	All node pools use x86-64 VM sizes (for example, Dsv5 or Esv5 families). No Arm-based (Ampere) sizes.	[R22]
At least two worker nodes with 32 GB RAM	Pega node pool minimum of three nodes (one per zone), each 64 GiB or more.	[R22]
Database in the same region as workers	Database and AKS in the same Azure region; zone-redundant database option.	[R22]
Tomcat from the Pega image	Only Pega-provided platform/pega, platform/installer and SRS images, mirrored to Azure Container Registry.	[R22, R28]
Load balancer and TLS	Application Gateway WAF v2 with the ingress controller; TLS to the pod using tier.service.tls.	[R22, R28]
External services	Confluent Cloud (Kafka) and SRS with the search cluster. Cassandra only if CDH or Process AI is licensed.	[R22]

6.2 Node pools

Table 13. AKS node pools

Pool	Purpose	Sizing guidance	Scheduling
system	AKS system pods	Three nodes across zones; AKS recommended system pool size	CriticalAddonsOnly taint
pega	Web and batch tiers	Start from the Pega tier defaults: CPU request 3, limit 4; memory 12 GiB; heap 8192m [R28]. Size so that each node holds at least two Pega pods with headroom for a rolling update.	nodeSelector or affinity per tier; spread across zones
srs	SRS pods	SRS minimum for production: three instances, 2 CPU and 2 GB each [R16]	Can share a general user pool in non-production

Note. The node counts above are starting points. Final sizes come from the load test in Section 16. Pega recommends leaving memory-based HPA disabled (hpa.enableMemoryTarget false) and scales on CPU by default, with hpa.targetAverageCPUValue 2.55 [R28].

6.3 Network design

Table 14. Subnets

Subnet or zone	Contents	Notes
AKS node subnet	AKS nodes (and pods if Azure CNI Overlay is not used)	Size for maximum nodes during upgrades (surge).
Private endpoint subnet	Private endpoints for Confluent (one per zone), search service, database, Key Vault, ACR	Network policies for private endpoints enabled so NSGs apply.

Subnet or zone	Contents	Notes
Application Gateway subnet	Application Gateway WAF v2	Dedicated subnet, as required by Application Gateway.
Hub VNet (peered)	Azure Firewall, private DNS resolver, shared services	Default route from AKS to the firewall.

6.3.1 Required network flows

Table 15. Required network flows

From	To	Port	Protocol	Purpose
Users	Application Gateway	443	HTTPS	Pega UI and APIs
Application Gateway	Web tier pods	443 (tier TLS port)	HTTPS	Backend TLS
Pega pods	Confluent private endpoints	9092 (and 443 for Confluent REST if used)	SASL_SSL	Stream service [R38]
Pega pods	SRS service	SRS service port	HTTP(S) with OAuth bearer	Indexing and search
SRS pods	Search private endpoint	443 or provider port (often 9200)	HTTPS	Index storage and queries
Pega pods and installer	Database private endpoint	Engine port	TLS	Rules and data
Pega and SRS pods	OAuth token endpoint	443	HTTPS	Client credentials token (Section 8.8)
ESO pods	Key Vault private endpoint	443	HTTPS	Secret synchronization
AKS nodes	ACR private endpoint	443	HTTPS	Image pulls
Pega pods	Pega Diagnostic Center	443	HTTPS	Through the firewall, if PDC is used

6.3.2 Private DNS

Every private endpoint needs a DNS record that resolves to its private IP from inside AKS. Azure Private DNS zones are linked to the spoke VNet or served by the hub DNS resolver [\[R46\]](#). Confluent Private Link needs the bootstrap record and one record per zone, as Confluent describes for Azure [\[R38\]](#). Missing zonal records are the most common cause of Kafka clients connecting to the bootstrap server and then failing on broker connections (Section 17).

Table 16. Private DNS zones

Service	DNS zone	Records
Confluent Cloud	The Confluent network DNS domain shown in the Confluent Cloud console	Wildcard and zonal records pointing at the private endpoint IPs, as listed in the Confluent console [R38]
Key Vault	privatelink.vaultcore.azure.net	Vault name A record [R46]
Container Registry	privatelink.azurecr.io	Registry and data endpoint records [R46]
Database	Engine-specific privatelink zone	Server A record [R46]
Search service	Provider-specific zone	As documented by the chosen provider (OD-04)

6.4 Images and registry

1. Download platform/installer, platform/pega and platform-services/search-n-reporting-service (or the -os variant) for each release from Pega Digital Software Delivery [\[R33\]](#).

2. Push them to Azure Container Registry under a repository per release, and scan them with the customer's image scanner.
3. Reference images by tag and record the digest in Appendix A, so that every environment runs the same build.
4. Grant AKS pull access through the kubelet managed identity (AcrPull role). Do not store registry passwords in values files.

6.5 Availability settings

- Spread the web and batch tiers across three zones with topology spread constraints or pod anti-affinity.
- Enable a PodDisruptionBudget on each tier (pdb.enabled true; minAvailable 1 by default) so node upgrades cannot remove every pod of a tier [\[R28\]](#).
- Keep at least two web pods and two batch pods in production. Queue processors and job schedulers run on the batch tier and stop if no batch pod is available.
- Use the default Pega readiness and liveness probes from the Helm chart unless a load test shows a need to change them.

7 Externalized Kafka on Confluent Cloud

7.1 What Pega uses Kafka for

Pega uses Kafka in two different ways, and they must not be confused. The first is the stream service: every queue processor and job scheduler uses it, so Pega is not fully functional without it [R13]. This is what the Helm stream section configures and what this section designs. The second is the Kafka data set, an optional application feature used mainly by Pega Customer Decision Hub and Process AI. Pega requires that Kafka data sets use a different Kafka service from the stream service [R13].

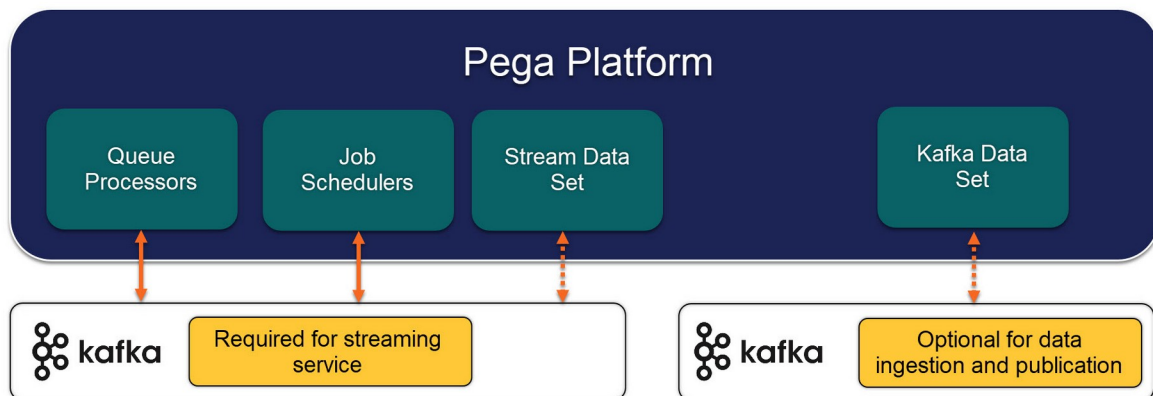


Figure 7. The two Kafka use cases in Pega Platform

Source: Pega Documentation, "External Kafka in your deployment" [R13]. © Pegasystems Inc. Reproduced with attribution.

7.2 Pega requirements mapped to Confluent Cloud

Confluent Cloud is a managed service, so some Kafka broker settings that Pega lists cannot be set by the customer. The table shows each requirement, how Confluent Cloud handles it, and what this runbook does about it.

Table 17. Pega Kafka requirements mapped to Confluent Cloud

Pega requirement [R13]	Confluent Cloud position	Action in this runbook
Kafka version compatible with the Pega client (4.0.0 in '26) [R3]	Fully managed; Confluent runs current Kafka versions.	Validate produce, consume and transactions from a Pega pod in every environment (V-K-05).
Confluent 5.4.x or later	Confluent Cloud is a current release line.	None beyond V-K-05.
message.max.bytes = 5000000	Topic max.message.bytes defaults to 2,097,164; editable; maximum 20,971,520 on Dedicated and Enterprise, 8,388,608 on Basic and Standard [R39].	After each start, check max.message.bytes on every Pega topic and raise it to 5,000,000 where lower (V-K-08). Repeat after Pega creates new topics.
replica.fetch.max.bytes, replica.fetch.response.max.bytes = 5000100	Broker replication settings are managed by Confluent and are not in the list of editable cluster settings [R40].	Logged in Section 23 (RC-04). Confirm with Confluent support that topics at 5,000,000 bytes replicate normally.
unclean.leader.election.enable = false	Managed by Confluent; not customer-editable [R40].	Logged in Section 23 (RC-04). Request written confirmation from Confluent.
auto.create.topics.enable = false	Default false on Dedicated; editable [R40].	Keep false. Pega creates topics through the Kafka AdminClient, which needs the CREATE operation in the topic ACL.
Generic Kafka security only; no IAM roles	SASL_SSL with API keys (PLAIN) is generic Kafka.	Use SASL_SSL with PLAIN (Section 7.5).
ACLs for TOPIC, GROUP, TRANSACTIONAL_ID and CLUSTER IDEMPOTENT_WRITE	Supported through Confluent ACLs on service accounts.	Create the ACLs in Section 7.5 before the first Pega start.

Pega requirement [R13]	Confluent Cloud position	Action in this runbook
Sizing: production 3 brokers, 4 cores, 16 GB, 200 GB; supports about 1000 partitions	Capacity is bought in eCKU or CKU; partition limits per unit are published [R37].	Size by partition count and throughput measured in rehearsal (Section 7.6).

7.3 Cluster type

Confluent offers Basic, Standard, Enterprise, Dedicated and Freight clusters [R37]. Basic and Standard are fine for short-lived development work but do not offer private networking. Freight does not support idempotent producers or transactions [R37], and Pega needs both: its ACL profile asks for IDEMPOTENT_WRITE and TRANSACTIONAL_ID permissions [R13]. That leaves Enterprise and Dedicated for any environment that holds real data.

Table 18. Confluent Cloud cluster limits relevant to Pega (source [R37])

Dimension (per unit)	Basic eCKU	Standard eCKU	Enterprise eCKU	Dedicated CKU
Ingress MBps	5	25	60	60
Egress MBps	15	75	180	180
Partitions (before replication)	30	250	3,000	4,500
Private networking	No	No	Yes (Private Link)	Yes (Private Link or VNet peering)
Fit for Pega production	No	No	Yes	Yes

Caution. Enterprise clusters that use Private Link on Azure are limited to 32 eCKU [R37]. Count the partitions for every environment that will share a cluster, including the about 100 extra partitions that Hazelcast removal adds per environment [R6], before choosing Enterprise.

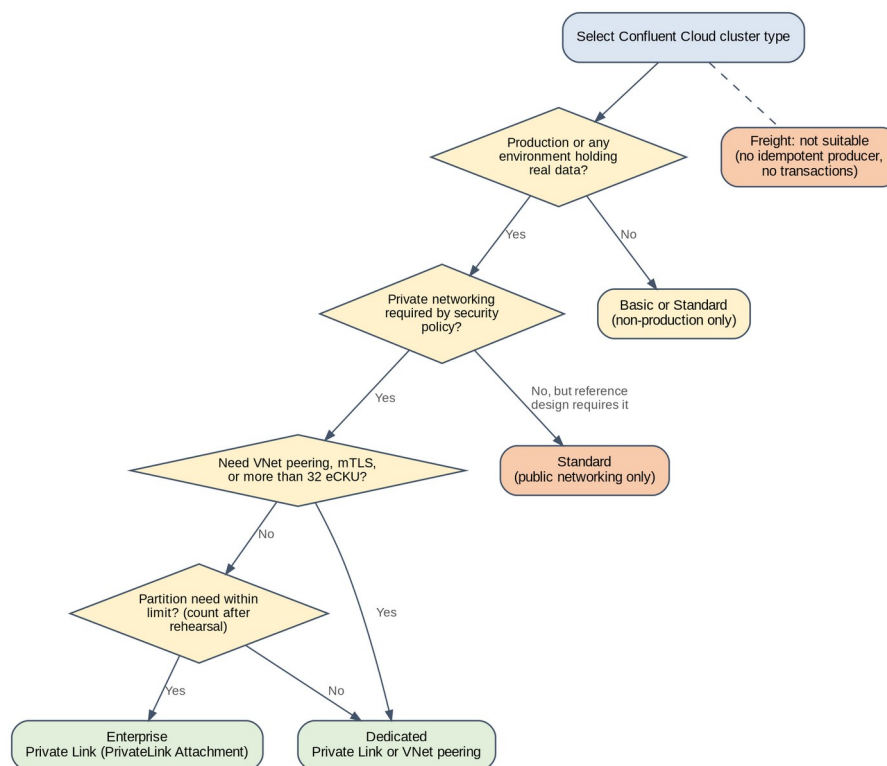


Figure 8. Decision: Confluent Cloud cluster type

Prepared for this implementation. Not a Pega-published diagram.

7.4 Private networking

Azure Private Link gives one-way private access from the customer VNet to Confluent Cloud. Confluent publishes the Azure set-up for Dedicated clusters: create or identify a Confluent network, add a Private Link access, create one private endpoint per zone in the customer subscription, set up DNS records, and validate connectivity [R38]. Enterprise clusters use Confluent's Private Link for serverless products, which follows the same customer-side pattern of private endpoints and DNS.

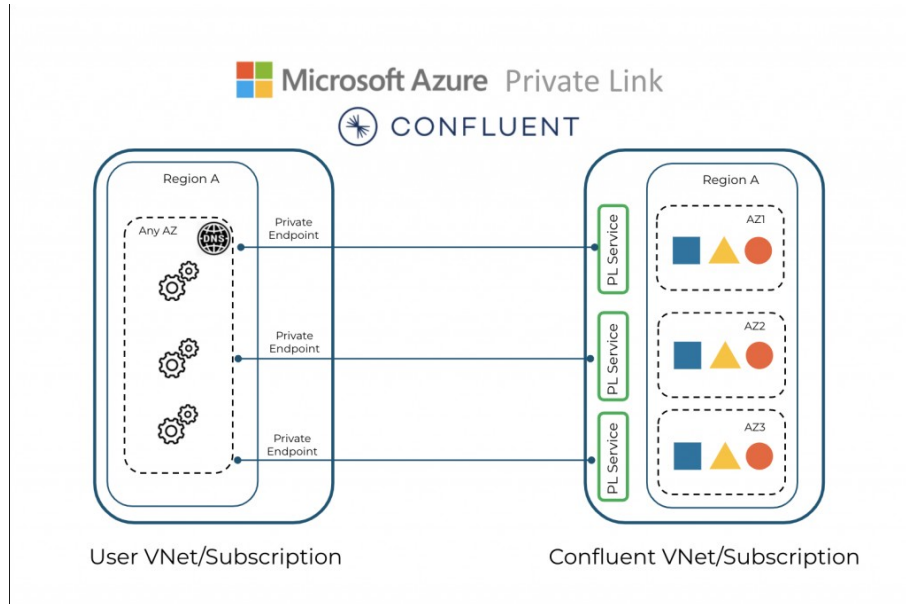


Figure 9. Azure Private Link between a customer VNet and Confluent Cloud

Source: Confluent Documentation, "Use Azure Private Link for Dedicated clusters on Confluent Cloud" [R38]. © Confluent, Inc. Reproduced with attribution.

+ Record set → Move ▾ Delete zone Refresh			
Essentials Resource group (move) : akumar-privatelink-test Subscription (move) : Cloud Private Development Subscription ID : d2b206bc-847a-437e-8d8d-aedd84a69ed0 Tags (edit) : Add tags			
ⓘ You can search for record sets that have been loaded on this page. If you don't see what you're looking for, you can try scrolling to allow more record sets to load.			
Search record sets			
Name	Type	TTL	Value
*	A	60	172.16.128.4 172.16.128.5 172.16.128.6
@	SOA	3600	Email: azureprivatedns-host.microsoft.com Host: azureprivatedns.net Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 10 Serial number: 1
*.az1	A	3600	172.16.128.4
*.az2	A	3600	172.16.128.5
*.az3	A	60	172.16.128.6

Figure 10. DNS records that Confluent lists for Azure Private Link

Source: Confluent Documentation [R38]. © Confluent, Inc. Reproduced with attribution.

Table 19. Private Link setup steps

Step	Action	Evidence to keep
N-1	Create the Confluent network in the same Azure region as AKS and register the customer subscription for Private Link.	Confluent network ID; subscription ID registered.

Step	Action	Evidence to keep
N-2	Create one private endpoint per zone in the private endpoint subnet, using the service aliases shown by Confluent.	Endpoint names, private IPs, connection state Approved.
N-3	Create the private DNS zone for the Confluent DNS domain and the wildcard and zonal records shown by Confluent. Link it to the spoke VNet or the hub resolver.	Zone records export.
N-4	From a pod in the Pega namespace, resolve the bootstrap name and a broker name; both must return private IPs.	nslookup output (Appendix D, command K-1).
N-5	From the same pod, open TLS to the bootstrap name on port 9092.	openssl s_client output showing the certificate chain (K-2).
N-6	Run a metadata request and confirm every broker name resolves privately.	kcat -L output (K-3).

Note. Confluent states that the Confluent CLI and Schema Registry access might need internet connectivity even when the cluster uses Private Link [R38]. Allow the Confluent control plane through the firewall for the administration jump host only, not for Pega pods.

7.5 Security: authentication and ACLs

7.5.1 Authentication choice

Pega lists four SASL options [R13], but the Helm chart's saslMechanism parameter accepts PLAIN, SCRAM-SHA-256 and SCRAM-SHA-512 [R28]. Confluent Cloud API keys use SASL/PLAIN. OAUTHBEARER has its own Pega article, but this design uses it only after it has been proven with Confluent in a rehearsal (Figure 11).

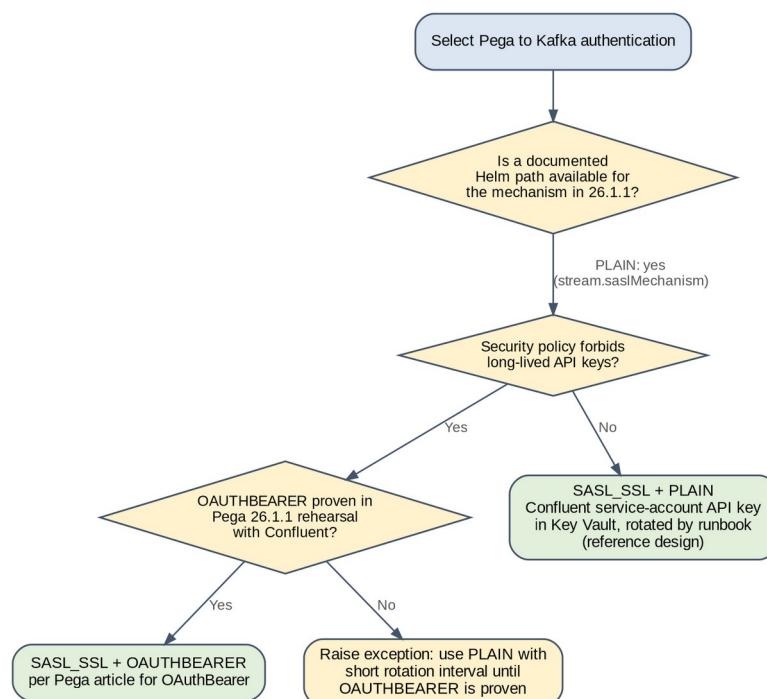


Figure 11. Decision: Pega to Kafka authentication

Prepared for this implementation. Not a Pega-published diagram.

7.5.2 Service account and ACLs

Create one Confluent service account per Pega environment. Grant it only the ACLs Pega lists, scoped to that environment's topic prefix. With streamNamePattern `pega-<env>-{stream.name}`, the prefix is `pega-<env>-`.

Table 20. Kafka ACLs for the Pega service account (source [R13])

Resource type	Resource name	Operation	Pattern	Why
TOPIC	<code>pega-<env>-</code>	ALL	PREFIXED	Create, write, read and delete Pega topics
GROUP	<code>pega-<env>-</code>	ALL	PREFIXED	Consumer groups for queue processors and data flows
TRANSACTIONAL_ID	*	READ, WRITE	LITERAL	Transactional producers
CLUSTER	the cluster	IDEMPOTENT_WRITE	LITERAL	Idempotent producers

Creating the ACLs

```
# Example with the Confluent CLI.
# Check flag names with: confluent kafka acl create --help
confluent kafka acl create --allow --service-account <sa-id> --operations all \
  --topic "pega-<env>-" --prefix --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> --operations all \
  --consumer-group "pega-<env>-" --prefix --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> --operations read,write \
  --transactional-id "*" --cluster <lkc-id>
confluent kafka acl create --allow --service-account <sa-id> \
  --operations idempotent-write --cluster-scope --cluster <lkc-id>
```

Important. The Hazelcast removal prerequisites name five topics (deployment.registry, NeoRemoteExecutionRequest, NeoRemoteExecutionResponse, SystemPulse_SystemPulseTopicName, Notification_Manager) [R6]. The documentation does not say whether streamNamePattern is applied to them. In the first rehearsal, list the topics Pega creates and confirm every one is covered by the prefixed ACL (V-K-07). If any is not, add a literal ACL for it.

7.6 Topic design and partition budget

Table 21. Topic design settings

Setting	Value	Reason
streamNamePattern	<code>pega-<env>-{stream.name}</code> ; a new <env> value in Release 2 (for example <code>prd2</code>)	Separates environments on a shared cluster; a new prefix in Release 2 keeps the 8.8 topics untouched for rollback (AD-05). Default is <code>pega-{stream.name}</code> [R28].
replicationFactor	3	Pega recommended value; cannot exceed the number of brokers [R28].
Topic creation	By Pega through AdminClient (CREATE in ACL)	Pega says <code>auto.create.topics.enable</code> is a legacy method [R13]. If security policy forbids CREATE, pre-create topics from the rehearsal inventory and the lists in [R3, R6].
max.message.bytes	5,000,000 on every Pega topic	Matches Pega's maximum message size [R13]; Confluent default is lower [R39].
SystemPulse_SystemPulseTopicName	6 partitions for '26	Pega '26 prerequisite; automatic if dynamic creation is enabled [R3].

Partition count drives both Confluent cost and limits. Measure it rather than estimate it:

1. After the first full start of each release in staging, count partitions under the environment prefix (Appendix D, command K-5).
2. Add about 100 partitions per environment for the Hazelcast removal messaging topics if they are not yet present [R6].
3. During Release 2, the 8.8 prefix and the new prefix exist together. Budget for both until the 8.8 topics are deleted after the rollback window closes.
4. Compare the total with the per-unit partition limit for the chosen cluster type (Table 18) and keep at least 30 % headroom for new queue processors.

7.7 Helm configuration of the stream service

The stream section of the Pega Helm chart connects every tier to Kafka [R14]. Plain-text passwords and the JAAS string are left empty and supplied through an external secret, whose keys must be STREAM_TRUSTSTORE_PASSWORD, STREAM_KEYSTORE_PASSWORD and STREAM_JAAS_CONFIG [R28].

values-<env>.yaml: stream section

```
stream:
  enabled: true
  bootstrapServer: "<bootstrap-host>:9092"
  securityProtocol: SASL_SSL
  saslMechanism: PLAIN
  trustStore: "" # set only if a private CA must be trusted
  keyStore: "" # not used with SASL/PLAIN
  jaasConfig: "" # supplied by the external secret
  streamNamePattern: "pega-<env>-{stream.name}"
  replicationFactor: "3"
  external_secret_name: "pega-stream-secret"
```

Value stored in Key Vault for STREAM_JAAS_CONFIG (stored as one line)

```
org.apache.kafka.common.security.plain.PlainLoginModule required
  username="<api-key>" password="<api-secret>";
```

If an application needs messages larger than 5,000,000 bytes, Pega requires the JVM arguments -Dstream.producer.max.request.size and -Dstream.producer.buffer.memory on every tier [R13]. They are set in catalinaOpts per tier [R28], and the topic max.message.bytes must be raised to match, within the Confluent maximum [R39].

7.8 prpcUtils after Hazelcast removal

From '25, prpcUtils needs its own connection to Kafka for clustering. Pega documents a -DNodeSettings argument in custom.jvm.args of prpcUtils.properties, and says this approach applies to '25.x and '26.x [R10]. Use it for any prpcUtils runs on 26.1.1, for example rule imports in a pipeline.

prpcUtils.properties (written as one line in the file)

```
custom.jvm.args=-Xmx4g -DNodeSettings=services/stream/provider=ExternalKafka;\
services/stream/broker/url=<bootstrap-host>:9092;\
services/stream/encryption/security/protocol=SASL_SSL;\
services/stream/encryption/sasl/mechanism=PLAIN;\
services/stream/encryption/sasl/jaas/config=<jas-from-key-vault>;\
services/stream/name/pattern=pega-<env>-;\
services/stream/external/replication/factor=3;\
dsm/services/stream/clustername=<lk-id>
```

Caution. This file holds the JAAS secret in plain text. Generate it at run time in the pipeline from Key Vault and delete it after the run. Pega recommends a heap of at least -Xmx4g [R10].

8 Externalized search: SRS with Elasticsearch or OpenSearch

8.1 How Pega search works from '24.2

SRS is a Pega backing service that sits between Pega Platform and the search cluster. Pega nodes send indexing and query requests to SRS, and SRS manages the indexes in Elasticsearch or OpenSearch [R16]. Embedded search and the legacy external plug-in were deprecated in 8.8 and are not available from '24.2 [R16]. SRS isolates the data of each Pega environment with a CUSTOMER_DEPLOYMENT_ID that becomes the index prefix [R16, R28].

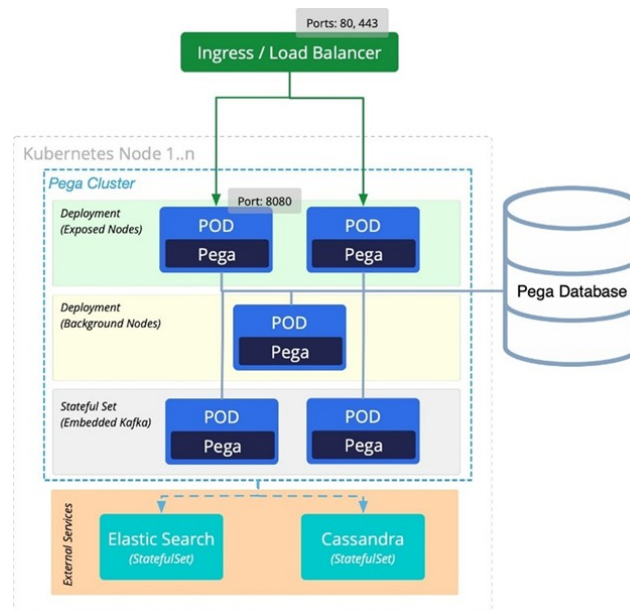


Figure 12. Earlier Pega cluster layout with embedded Kafka and in-cluster Elasticsearch

Source: Pega Academy, "Cloud deployment architecture" [R36]. © Pegasystems Inc. Reproduced with attribution. Shown for comparison with the 8.8 source; this layout is not supported from '24.2.

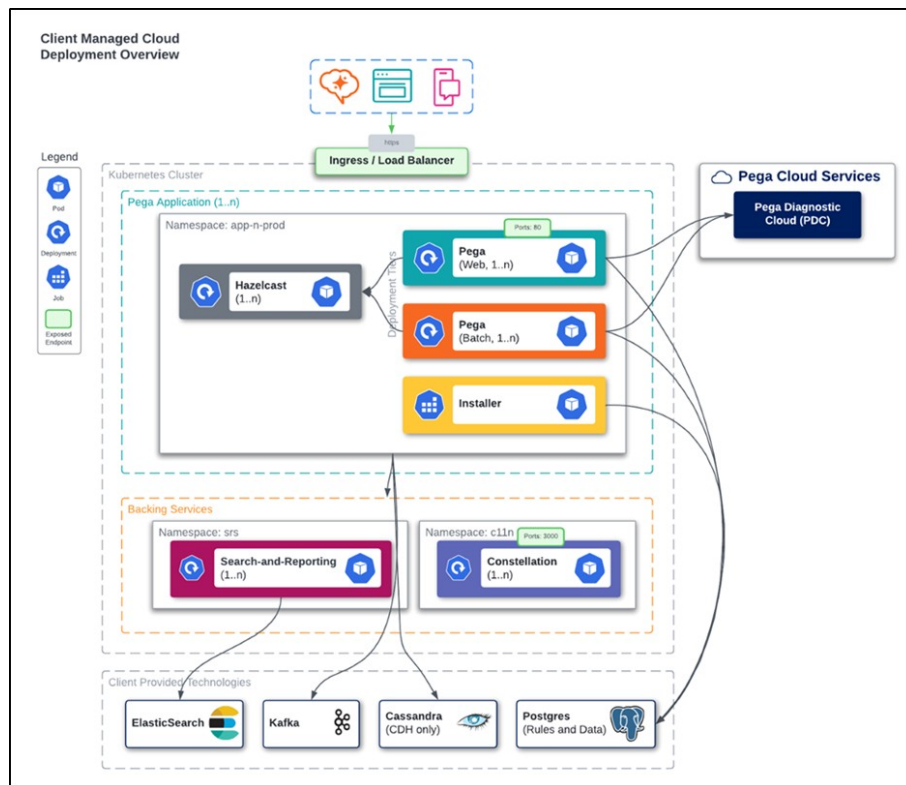


Figure 13. Client-managed cloud deployment overview with SRS as a backing service

Source: Pega Academy, "Search and Reporting Service" [R34]. © Pegasystems Inc. Reproduced with attribution. The Hazelcast pod in this illustration does not apply to '25 and later.

8.2 Requirements and supported versions

Table 22. SRS compatibility for Pega 8.6 and later, SRS 1.44.3 or later (source [R16])

SRS image	Authentication	Certified search versions ('26 documentation)	Pega best practice
search-n-reporting-service	Not enabled	Elasticsearch 7.17.9, 7.17.29	Not for production
search-n-reporting-service	Enabled	Elasticsearch 7.17.9, 7.17.29, 8.10.3, 8.15.1, 8.15.5, 8.18.2, 8.18.3, 8.19.11	Elasticsearch 8.18.3
search-n-reporting-service-os	Enabled	AWS OpenSearch service with Elasticsearch 7.10; OpenSearch 1.3, 2.15, 2.19	OpenSearch 2.15

Note. The Helm chart README lists a newer SRS (1.49.2) and more Elasticsearch versions (up to 8.19.22) than the '26 documentation page. Section 23 (RC-01) records this. This design selects versions that appear in the Pega documentation matrix, and uses the newest SRS image that the Helm README lists for the chart version deployed.

- Use only official Elasticsearch or OpenSearch images; custom images such as bitnami/elasticsearch are not supported [R16].
- Do not share the search cluster with non-Pega software [R16].
- The search cluster deployed by the backingservices chart is for development and test only; OpenSearch cannot be used for that option [R16, R29].
- SRS supports Pega 8.6 and later, so the same SRS deployment serves 8.8 in Release 1, the bridge release in Release 2 and 26.1.1 in Release 3 [R16].

8.3 Choosing the search provider

Fixed decision D-08 excludes an Azure-managed Elasticsearch service. Azure does not offer a first-party managed OpenSearch service either, so the provider is a third party or the customer. Pega accepts open-source, licensed or cloud subscription options [R16].

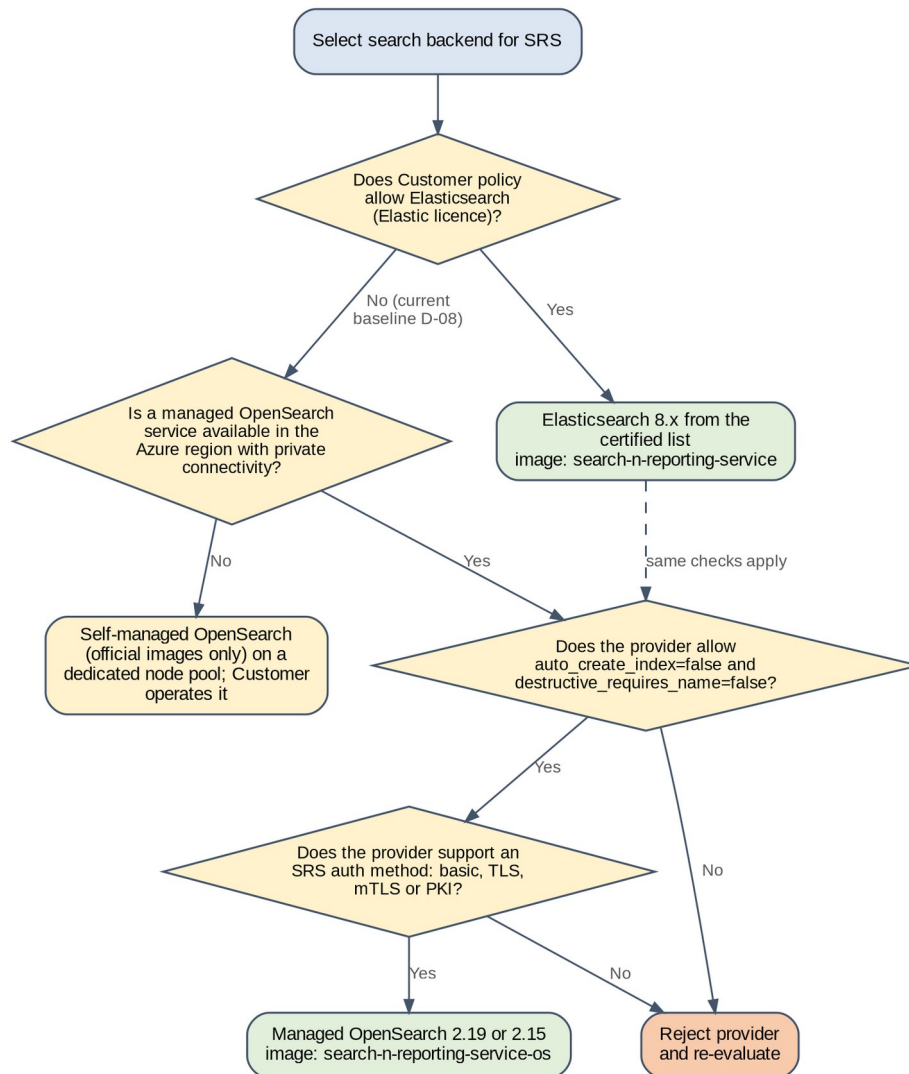


Figure 14. Decision: search backend for SRS

Prepared for this implementation. Not a Pega-published diagram.

Table 23. Search provider options

Option	SRS image	Strengths	Points to validate
Elastic Cloud hosted on Azure	search-n-reporting-service	Vendor-managed; Elasticsearch versions on the Pega matrix	Elastic licence acceptance; private connectivity from the customer VNet; version pinning to a matrix version
Managed OpenSearch from a third-party provider running on Azure	search-n-reporting-service-os	Apache 2.0 licence; vendor-managed	Region availability; private connectivity; ability to set action.auto_create_index and action.destructive_requires_name; version 2.15 or 2.19 available
Self-managed OpenSearch on a dedicated AKS node pool	search-n-reporting-service-os	Full control; no third-party contract	Customer operates upgrades, backups, scaling and on-call; official images only

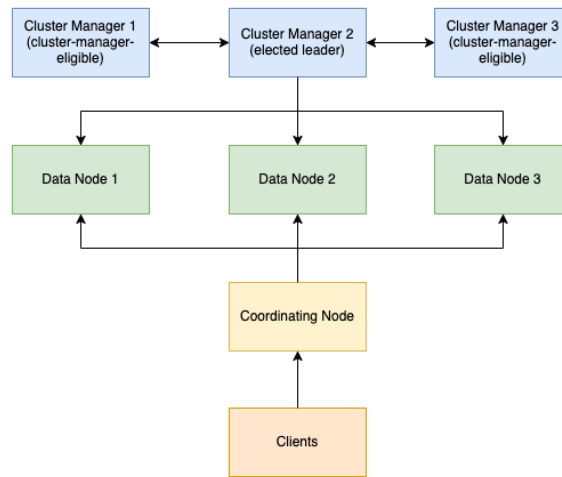


Figure 15. OpenSearch cluster roles: cluster-manager and data nodes

Source: OpenSearch Documentation, "Creating a cluster" [R42]. © OpenSearch contributors. Reproduced with attribution.

8.4 Sizing

Pega publishes three sizing examples. Start from the default table for production and adjust after measuring indexing and query performance, as Pega advises [R16].

Table 24. Pega default sizing for search and SRS (source [R16])

Service	Landscape	Instances	CPU each	RAM GB each	Storage GB each
Cluster-manager (master) nodes	Production, stage	3	2	8	N/A
Data nodes	Production, stage	3	4	16	100
Data nodes	Testing, development	1	2	8	100
SRS	Production, stage	3 (autoscaled)	2	2	N/A
SRS	Testing, development	1 (autoscaled)	2	2	N/A

Table 25. Pega sizing example for about 2 TB of searchable data and about 750,000 documents (source [R16])

Service	Landscape	Instances	CPU each	RAM GB each	Storage GB each
Cluster-manager nodes	Production, stage	3	2	8	N/A
Data nodes	Production, stage	9	4	16	250
Data nodes	Testing, development	4	4	16	250
SRS	Production, stage	3 (autoscaled)	2	2	N/A

Pega's minimum production sizing is three cluster-manager nodes (2 CPU, 4 GB) and three data nodes (2 CPU, 8 GB, 50 GB) [R16]. Release 2 builds a second set of indexes under a new customerDeploymentId while the 8.8 indexes still exist, so plan data node storage for two copies of the production indexes until the rollback window closes.

8.5 Search cluster settings

SRS needs two cluster settings. If the SRS user has the manage cluster privilege, SRS disables index auto-creation itself; otherwise set it manually. Elasticsearch 8 and later also need `destructive_requires_name` set to false because SRS deletes indexes with a pattern [\[R16\]](#).

Cluster settings required by SRS

```
PUT _cluster/settings
{
  "persistent": {
    "action": {
      "auto_create_index": "false",
      "destructive_requires_name": "false"
    }
  }
}
```

Caution. The setting `destructive_requires_name` false allows index deletion by wildcard for every user with delete rights. Limit index delete privileges to the SRS user, and restrict administrator access to the search cluster to a break-glass group.

8.6 SRS deployment (backingservices chart)

backingservices values-<env>.yaml (OpenSearch example)

```
global:
  k8sProvider: "aks"
  imageCredentials:
    registry: "<acr-name>.azurecr.io"
srs:
  enabled: true
  deploymentName: "srs-<env>"
  srsRuntime:
    replicaCount: 3
    # -os image for OpenSearch; search-n-reporting-service for Elasticsearch
    srsImage: "<acr-name>.azurecr.io/platform-services/search-n-reporting-service-os:<tag>"
    env:
      AuthEnabled: true
      OAuthPublicKeyURL: "<idp-jwks-url>"
    ssl:
      enabled: true
      clientAuthentication: "want"      # "need" enforces mTLS from Pega
      keystore:
        file: "srs-keystore.p12"
        type: "PKCS12"
      truststore:
        file: "srs-truststore.jks"
        type: "JKS"
      certsSecret: "srs-runtime-certs" # store files and their passwords
  srsStorage:
    provisionInternalEScluster: false
    domain: "<search-host>"
    port: 443
    protocol: https
    tls:
      enabled: true
    # certificateName and certsSecret are set only if the provider uses a private CA
    basicAuthentication:
      enabled: true
    authSecret: "srs-search-credentials" # keys: username, password
    requireInternetAccess: false
    networkPolicy:
      enabled: true
```

The keys come from the SRS chart README [\[R29\]](#). Use `search-n-reporting-service` instead of the `-os` image if the backend is Elasticsearch. The README also documents mTLS and PKI client-certificate options between SRS and the search cluster, and make targets (`external-es-secrets`, `update-external-es-secrets`, `srs-mtls-prerequisite`) that create the certificate secrets [\[R29\]](#).

8.7 Connecting Pega to SRS

The pegasearch section of the Pega Helm chart points Pega at SRS. Remove the legacy parameters (image, memLimit, replicas) that described the old plug-in [R18].

values-<env>.yaml: Pega to SRS

```
global:
  customerDeploymentId: "<env-deployment-id>" # index prefix; equals the guid claim
pegasearch:
  externalSearchService: true
  externalURL: "https://srs-<env>.srs-<env>.svc.cluster.local"
  srsAuth:
    enabled: true
    url: "<idp-token-endpoint>"
    clientId: "<srs-client-id>"
    scopes: "pega.search:full"
    authType: "private_key_jwt"
    privateKeyAlgorithm: "RS256"
    external_secret_name: "pega-srs-oauth" # key: SRS_OAUTH_PRIVATE_KEY
  srsMTLS:
    enabled: false # true if SRS requires client certificates
    external_secret_name:
```

8.8 Authorization between Pega and SRS

Pega recommends an OAuth service of the customer's choice to authorize Pega to SRS [R16]. Pega obtains a token with the client credentials grant, using a private key (private_key_jwt) or a client secret (client_secret_basic), and the scope pega.search:full [R18, R28]. SRS validates the token signature with the public key URL and checks that the guid claim equals the customerDeploymentId [R28].

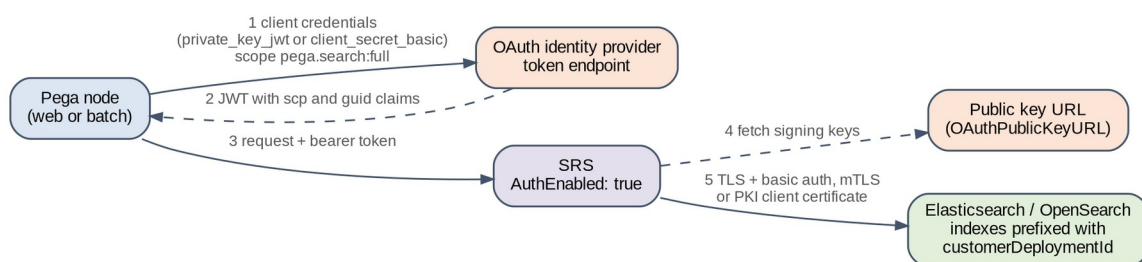


Figure 16. Token flow between Pega, the identity provider and SRS

Prepared for this implementation. Not a Pega-published diagram.

Important. Open decision OD-16: confirm that the chosen identity provider can issue the claims SRS checks. For example, Microsoft Entra ID access tokens for the client credentials flow carry application permissions in the roles claim rather than the scp claim [R45], and do not carry a guid claim by default.

Prove the token in the first non-production environment by decoding it and confirming the scope and guid values (V-S-06) before any production build depends on it.

8.9 Customer deployment ID

- Set global.customerDeploymentId explicitly in every environment. If left empty, it defaults to the namespace name [R28].
- Treat it as immutable for the life of an environment. Pega asks for an immutable CUSTOMER_DEPLOYMENT_ID when several environments share one SRS [R17].
- Use a new value in Release 2 (for example <env>-r2) so that the bridge release builds new indexes and the 8.8 indexes remain for rollback (AD-06).
- Keep the Release 2 value unchanged in Release 3, so the zero-downtime update keeps the same indexes.

8.10 Encryption between Pega and SRS

Pega's best practice is to encrypt Pega-to-SRS traffic at the infrastructure layer with a service mesh, or to use TLS [\[R16\]](#). This design enables TLS on SRS (`srsRuntime.ssl.enabled`) and uses the https SRS URL. Setting `clientAuthentication` to `need` enforces mutual TLS, which requires `pegasearch.srsMTLS` with the Pega keystore and truststore [\[R28, R29\]](#).

9 Secrets, certificates and identity

Fixed decision D-09 forbids plain-text secrets in Helm values. The Pega Helm chart supports the External Secrets Operator for database, registry, stream, SRS and certificate secrets [R28]. Key Vault is the system of record.

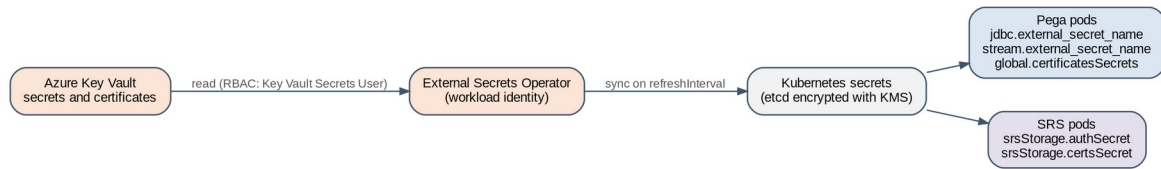


Figure 17. Secret delivery from Key Vault to Pega and SRS pods

Prepared for this implementation. Not a Pega-published diagram.

Table 26. Secret inventory

Kubernetes secret	Keys	Consumed by	Key Vault secret	Rotation
pega-db-secret	DB_USERNAME, DB_PASSWORD (or token-based settings)	Pega tiers, installer (jdbc.external_secret_name)	<env>-pega-db	Per database policy
pega-stream-secret	STREAM_TRUSTSTORE_PASSWORD, STREAM_KEYSTORE_PASSWORD, STREAM_JAAS_CONFIG	Pega tiers (stream.external_secret_name)	<env>-confluent-jaas	API key rotation, Section 9.1
pega-srs-oauth	SRS_OAUTH_PRIVATE_KEY	Pega tiers (pegasearch.srsAuth.external_secret_name)	<env>-srs-oauth-key	Yearly or per IdP policy
srs-search-credentials	username, password	SRS (srsStorage.authSecret)	<env>-search-user	Quarterly
srs-runtime-certs	srs-keystore.p12, srs-truststore.jks, keystorePassword, truststorePassword	SRS (srsRuntime.ssl.certsSecret)	<env>-srs-tls	Before certificate expiry
pega-tier-tls	TOMCAT_KEYSTORE_CONTENT, TOMCAT_KEYSTORE_PASSWORD, ca.crt	Web tier TLS (tier.service.tls)	<env>-pega-tls	Before certificate expiry

Key names for the stream and SRS OAuth secrets are fixed by the Pega Helm chart [R28]; key names for the web tier keystore are also fixed [R28]. The database secret keys follow the chart's jdbc section. Secret names are this design's convention.

External Secrets Operator objects (Azure Key Vault provider [R47])

```

apiVersion: external-secrets.io/v1beta1
kind: SecretStore
metadata:
  name: keyvault
  namespace: pega-<env>
spec:
  provider:
    azurekv:
      authType: WorkloadIdentity
      vaultUrl: "https://<vault-name>.vault.azure.net"
      serviceAccountRef:
        name: eso-keyvault-reader
---
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: pega-stream-secret
  namespace: pega-<env>
spec:

```

```

refreshInterval: 1h
secretStoreRef:
  name: keyvault
  kind: SecretStore
target:
  name: pega-stream-secret
data:
  - secretKey: STREAM_TRUSTSTORE_PASSWORD
    remoteRef: { key: <env>-stream-truststore-password }
  - secretKey: STREAM_KEYSTORE_PASSWORD
    remoteRef: { key: <env>-stream-keystore-password }
  - secretKey: STREAM_JAAS_CONFIG
    remoteRef: { key: <env>-confluent-jaas }

```

Note. Check the External Secrets Operator API version installed in the cluster (v1beta1 or v1) and use it in these manifests. The truststore and keystore password keys can hold empty values when SASL/PLAIN is used without a private CA, but keep the keys so the order Pega expects is preserved [R28].

9.1 Rotating the Confluent API key

1. Create a second API key for the same service account in Confluent Cloud. The ACLs belong to the service account, so they apply to both keys.
2. Write the new JAAS string to the Key Vault secret `<env>-confluent-jaas` as a new version.
3. Wait for the External Secrets Operator refresh, or force it by annotating the ExternalSecret. Confirm the Kubernetes secret changed.
4. Run a rolling restart of the Pega tiers (kubectl rollout restart deployment). Pods read the JAAS value at start.
5. Confirm the Stream landing page shows Status NORMAL and queue processors are processing.
6. Delete the old API key in Confluent Cloud. Record the change in the operations log.

9.2 Certificates

Table 27. Certificates

Certificate	Issued by	Used for	Renewal owner
Public TLS certificate for the Pega host name	Customer public CA	Application Gateway listener	Platform team
Web tier backend certificate	Customer private CA	Application Gateway to web pods	Platform team
SRS server certificate	Customer private CA	Pega to SRS TLS	Platform team
Confluent broker certificates	Confluent (public CA)	SASL_SSL from Pega	Confluent
Search service certificate	Provider or customer CA	SRS to search cluster TLS	Search provider

Caution. Do not route Pega-to-Confluent or SRS-to-search traffic through a TLS-inspecting proxy. Re-signed certificates break the trust chain the clients check. Private endpoint traffic stays inside the VNet and does not need inspection.

10 Upgrade strategy from Pega 8.8 to 26.1.1

10.1 What changed between 8.8 and 26.1.1

Table 28. Changes between 8.8 and 26.1.1 that affect Kafka, search and the update path

Release	Change that affects this programme	Source
8.8	Embedded Kafka, Elasticsearch and Cassandra deprecated. VM-based deployment deprecated. Stream tier (nodeType Stream) deprecated.	[R2, R30]
'24.1	Last release that supports both embedded services and VM-based deployments.	[R2]
'24.2	External Kafka, external search and (for CDH and Process AI) external Cassandra required. Embedded Elasticsearch and the plug-in removed.	[R2, R16]
Patches 23.1.4, 24.1.3, 24.2.2 and later	Hazelcast removal tooling: HazelcastDecommission service package and the prconfig/cluster/hazelcast/disabled/default DSS.	[R5, R9]
'25	Containerized deployment only. Hazelcast removed. Tomcat 10.1 (Jakarta EE 9) only. All tables need primary keys.	[R2, R4]
'26	Kafka client 4.0.0 (backward compatible with 3.9.2). Partitioned, multi-threaded queue processing (PEGA0179). SystemPulse topic with 6 partitions. Cassandra 5.0 support.	[R3]

10.2 Why 8.8 cannot be updated directly

Pega requires Hazelcast to be removed before an update to '25 or later [\[R5\]](#). The removal needs functions that exist only in patches 23.1.4, 23.1.5, 24.1.3, 24.1.4, 24.2.2, 24.2.3 and later [\[R9\]](#). Pega 8.8 has neither the HazelcastDecommission service package nor the DSS behavior, so it cannot remove Hazelcast, and therefore cannot be updated straight to 26.1.1. The system must pass through a bridge release.

Evidence. Pega's page "Adopting Pega Platform infrastructure changes" gives the sequence for clients on older releases: update to the latest patch of '24.1, externalize Elasticsearch, Kafka and (if used) Cassandra, remove Hazelcast, shift from virtual machines to containers, then update to '25 [\[R2\]](#). The '26 prerequisites page repeats that Hazelcast must be removed before updating to '25 or later [\[R3\]](#).

10.3 Release plan

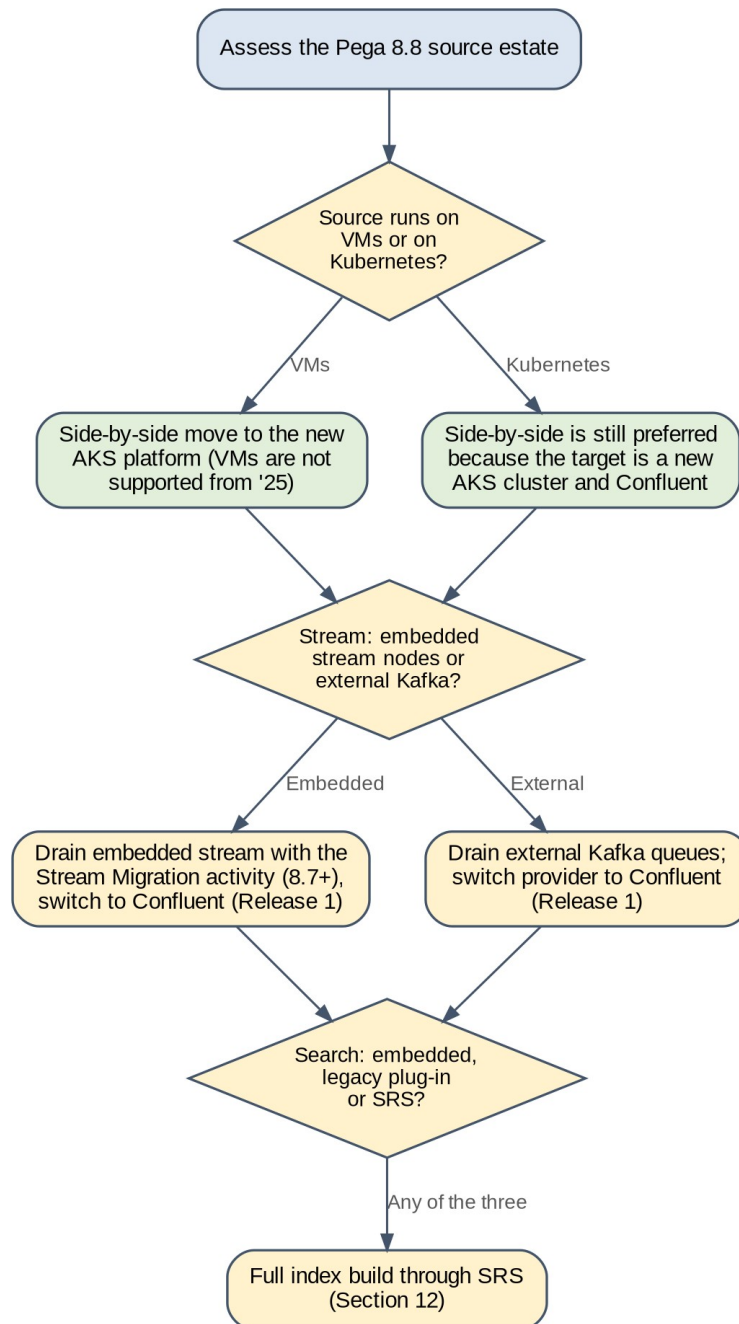


Figure 18. Assessing the 8.8 source for Release 1

Prepared for this implementation. Not a Pega-published diagram.

Table 29. What each release changes

	Release 1: externalize on 8.8	Release 2: platform move	Release 3: update to 26.1.1
Pega version	8.8 (unchanged)	8.8 to bridge ('24.1 latest patch)	Bridge to 26.1.1
Runtime	Current 8.8 servers	New AKS cluster	Same AKS cluster
Kafka	Current stream to Confluent (drain first)	Same Confluent cluster, new prefix	Same topics

	Release 1: externalize on 8.8	Release 2: platform move	Release 3: update to 26.1.1
Search	Current search to SRS (full build)	Same SRS, new customerDeploymentId (full build)	Same indexes
Hazelcast	Unchanged	Removed with the DSS method	Absent
Database	Unchanged	Copied to Azure and updated in place	Zero-downtime update (new rules schema)
Outage	Yes: drain, restart, index build	Yes: drain, copy, update, index build	Near zero (rolling restart)
Rollback	Point 8.8 back to the old stream and search	Keep 8.8 and its database; revert DNS	Restore pre-update backup (Section 15.4)

10.3.1 Choice of bridge release

The bridge release is a transit point, used only until Release 3. This design uses the latest '24.1 patch, at least 24.1.4, for three reasons. Pega names '24.1 in its adoption sequence [R2]. Patch 24.1.4 includes the Hazelcast removal changes without a hotfix, whereas 24.1.3 needs HFIX-C2252 [R7]. And '24.1 still accepts the embedded services the 8.8 estate may use, which keeps the rehearsal of Release 2 independent of Release 1 timing. Because Release 1 removes the embedded services first, the latest '24.2 patch (24.2.3 or later) is also a valid bridge; OD-02 records the choice and asks Pega Support to confirm the supported path from the selected bridge to 26.1.1.

10.3.2 Update method and staging

Pega's guidance is to update a staging environment first and to test there before production [R25, R26]. The zero-downtime checklist adds that copying production data into staging is not recommended; staging should hold appropriate test data [R25]. Rehearsal timings for drain, copy, update and index build only mean something if the volumes are close to production, so OD-13 asks the customer to decide between masked production-scale data and a synthetic data set of the same size.

10.4 Preparation on 8.8 (all releases)

Table 30. Preparation tasks

#	Task	How	Output	Needed before
P-1	Run the Pega Upgrade Tools against 8.8	Pega Upgrade Tools for the target release [R11]	List of impacted rules, CodeSets and JARs	Release 2 rehearsal
P-2	Prepare Jakarta EE versions of custom JARs	Rebuild against jakarta.* packages; Pega notes that not every javax.* package maps to jakarta.* [R11]	New JARs held in the artifact repository	Release 3 rehearsal
P-3	Set each impacted CodeSet DSS version to latest	Supported from 8.8 [R11]	DSS change record	Release 3
P-4	Scan for missing primary keys	primaryKeyUtility dry run; check database free space, because the utility copies tables [R12]	Report of tables without primary keys	Release 2 rehearsal
P-5	Create missing primary keys	primaryKeyUtility (non-dry run) in a maintenance window	Utility log	Release 3
P-6	Review custom queue processors	Inventory and test plan for the '26 partitioned processing model; remove DSS delayeditems/dataflowbased/threadspn node [R3]	Queue processor test cases	Release 3 rehearsal
P-7	Confirm database support	Platform Support Guide for 26.1.1 [R1]	Engine and version confirmed	Release 2 design

#	Task	How	Output	Needed before
P-8	Inventory Kafka data sets and integrations	Section 11.3	Integration list with owners	Release 1 design
P-9	Measure searchable data	Search landing page and database counts per indexed class	Baseline for sizing and V-S checks	Release 1 design
P-10	Resolve broken queue items	Admin Studio, queue processors, Broken items	Backlog near zero	Each cutover

10.5 Release 1: externalize Kafka and search on 8.8

Release 1 changes only the stream provider and the search provider. Pega's '26 Kafka and search pages have 8.8 versions (select 8.8 in the documentation version list), and SRS supports 8.6 and later [\[R13, R16\]](#). Section 11.6 and Section 12.6 hold the detailed steps; Figure 19 shows the order.

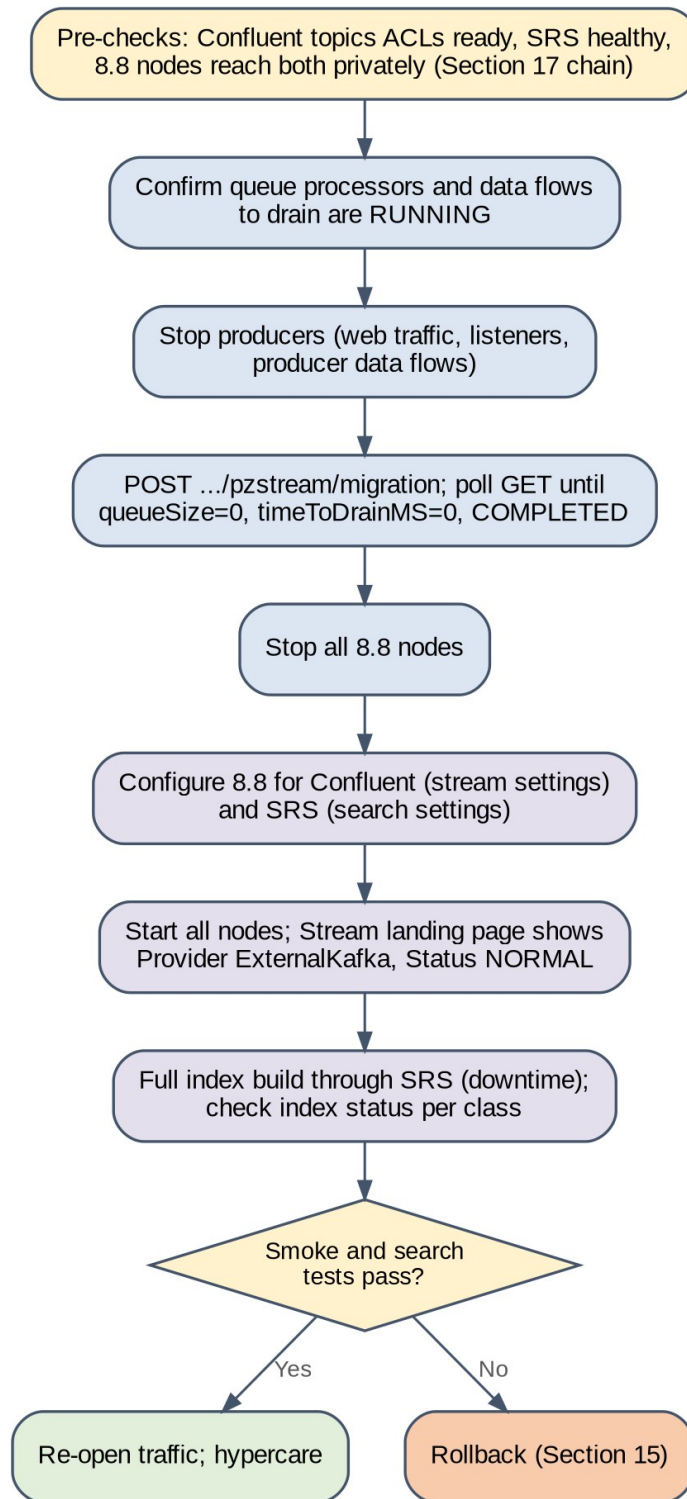


Figure 19. Release 1 sequence: drain, switch and rebuild on 8.8

Prepared for this implementation. Not a Pega-published diagram.

Note. If the 8.8 nodes run on VMs outside Azure, Release 1 needs private routing from those servers to the Confluent private endpoints and to SRS (for example ExpressRoute or VPN into the hub VNet). SRS has no ingress in the target design, so Release 1 needs an internal load balancer for SRS that only the 8.8 server subnets can reach. Remove it after Release 2.

10.6 Release 2: platform move to AKS on the bridge release

10.6.1 Sequence

1. Drain the 8.8 queues with the Stream Migration activity, then stop all 8.8 nodes (Section 11.6).
2. Take the final database backup and restore it to the Azure database.
3. Run the installer job with the bridge installer image, `global.actions.execute: upgrade` and `installer.upgrade.upgradeType: in-place` [R28].
4. Deploy the bridge tiers (`global.actions.execute: deploy`) with the new streamNamePattern prefix and the new customerDeploymentId. Use embedded Hazelcast settings (`hazelcast.enabled false`, `clusteringServiceEnabled false`) [R28].
5. Remove Hazelcast with the downtime method (Section 10.6.2).
6. Run the full index build through SRS (Section 12.6).
7. Run smoke tests, then switch DNS (Section 14).

10.6.2 Hazelcast removal with the DSS method

Before the window, confirm the Pega prerequisites: database CPU at least 10 % below its alert threshold, Kafka capacity for about 100 extra partitions for the environment, and Kafka CPU at least 15 % below its alert threshold [R6]. Pega also advises scaling down to the minimum number of nodes [R8].

Table 31. Hazelcast removal steps on the bridge release

Step	Action [R8]	Expected result
H-1	Dev Studio > Records > SysAdmin > Dynamic System Settings. Find <code>prconfig/cluster/hazelcast/disabled/default</code> . If it is missing, create it with owning ruleset Pega-Engine, type String, and select "Is restart required".	DSS exists
H-2	Set its value to true and save.	Value true
H-3	Stop all Pega nodes (scale every tier to 0 replicas).	No Pega pods running
H-4	Start all Pega nodes (restore replica counts).	All pods Ready
H-5	Skip Clustering Service removal: the bridge runs embedded Hazelcast. (If the Clustering Service was used, set <code>hazelcast.enabled</code> and <code>clusteringServiceEnabled</code> to false and run helm upgrade.)	No Clustering Service pods
H-6	Records > Integration-Resources > Service Package > HazelcastDecommission. Run <code>hazelcast/disabled</code> with GET.	Response true
H-7	Run <code>checkRemoteExecutionConnectivity</code> with GET.	Response <code>allNodesReachable</code> [R7]

10.7 Release 3: zero-downtime update to 26.1.1

Release 3 runs on the platform built in Release 2. The Pega Helm chart's zero-downtime upgrade type migrates the rules into a new rules schema, uses a temporary data schema, upgrades the rules, performs a rolling reboot onto the new rules, then upgrades the data schema [R28, R33]. It is supported for updates from 8.4.2 and later [R28]. Pega's update procedure for client-managed deployments has four parts: prepare the environment, update the database schemas, update the nodes with the Helm chart, and complete the post-update tasks [R27].

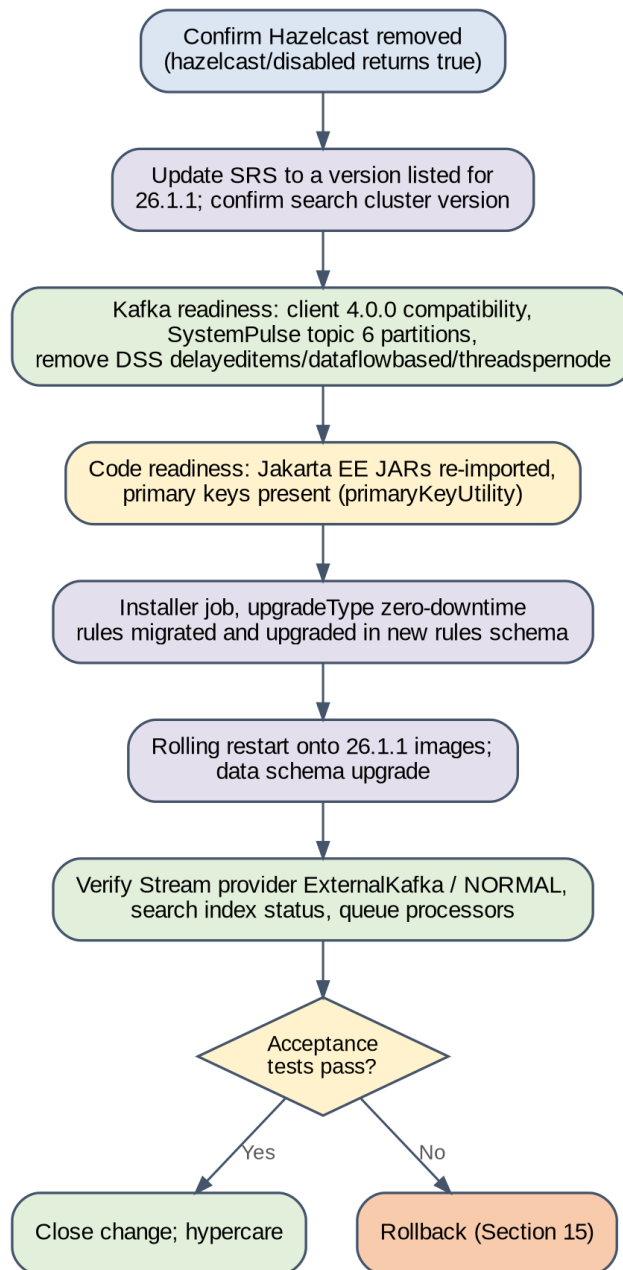


Figure 20. Release 3 sequence on the AKS platform

Prepared for this implementation. Not a Pega-published diagram.

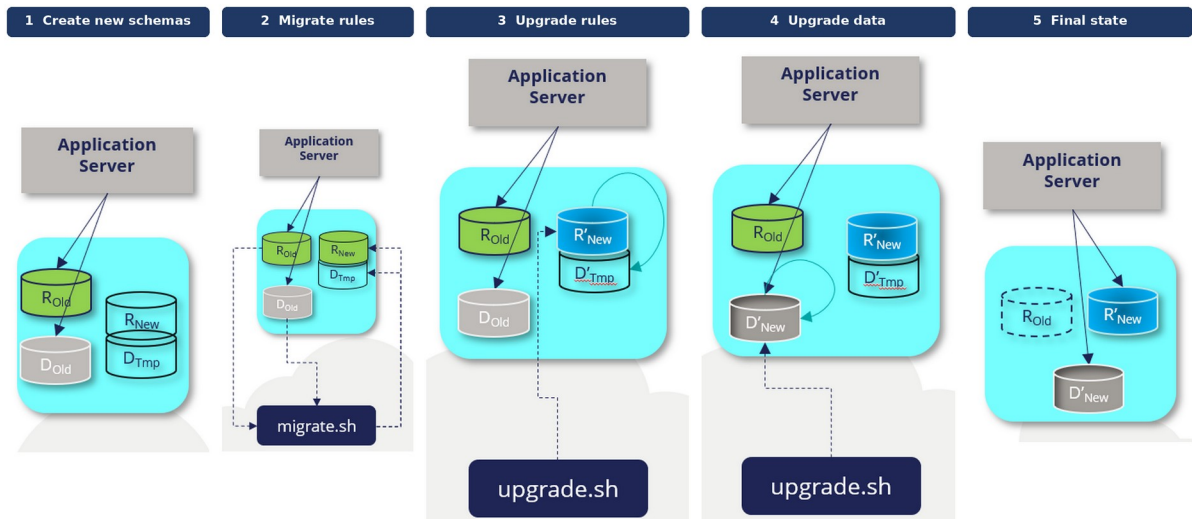


Figure 21. Schema states during a zero-downtime update

Source: Pega Helm charts, "Upgrading Pega Platform in your deployment with zero-downtime" [R33]. © Pegasystems Inc., Apache License 2.0. Images combined and numbered for this document.

values-<env>.yaml: installer section for Release 3 [R28]

```
global:
  actions:
    execute: "upgrade-deploy"
installer:
  image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
upgrade:
  upgradeType: "zero-downtime"
  targetRulesSchema: "<new-rules-schema>"
  targetDataSchema: "<temp-data-schema>"
```

Table 32. Release 3 steps

Step	Action	Source
U-1	Confirm Hazelcast removal (H-6 and H-7 results from Release 2).	[R7, R8]
U-2	Update SRS to the image version listed for the Helm chart that deploys 26.1.1; confirm the search cluster version is on the matrix.	[R16, R29]
U-3	Confirm Confluent compatibility with the Kafka 4.0.0 client in staging (V-K-05).	[R3]
U-4	If topic creation is restricted, set SystemPulse_SystemPulseTopicName to 6 partitions.	[R3]
U-5	Remove DSS delayeditems/dataflowbased/threadspernode; retest custom queue processors.	[R3]
U-6	Confirm primary keys exist on all tables (P-5).	[R12]
U-7	Update the Helm values: 26.1.1 images, installer section above. Run helm upgrade.	[R28, R33]
U-8	After the update, import the Jakarta EE JARs into a higher CodeSet version and restart.	[R11]
U-9	Configure prpcUtils for Kafka if pipelines use it.	[R10]
U-10	Run the post-update checks: Stream landing page, index status, queue processors, regression pack.	[R15, R21]
U-11	Drop the temporary data schema after the update, as the Helm README instructs.	[R28]

11 Kafka topic migration: requirements, validation, strategy and execution

11.1 Requirements

Table 33. Kafka migration requirements

ID	Requirement	Source
R-K-01	No message that is in transit in a Pega stream queue at the time of a provider switch may be lost.	[R15]
R-K-02	Every queue processor and data flow that holds messages must be RUNNING when the drain starts, because the Stream Migration activity does not drain STOPPED, FAILED or NOT RUNNING ones.	[R15]
R-K-03	The target Kafka must meet Pega's settings, ACLs and authentication rules before the first start.	[R13]
R-K-04	Pega must report Provider ExternalKafka and Status NORMAL on the Stream landing page after each start.	[R15]
R-K-05	Topic names must carry an environment-specific prefix so environments sharing a cluster cannot read each other's data.	[R13, R28]
R-K-06	Application Kafka integrations (Kafka data sets) must keep working, or be repointed with an agreed offset position.	[R13]
R-K-07	The source topics must remain intact until the rollback window for each release closes.	AD-05

11.2 What can and cannot be migrated

Pega states that existing Stream data cannot be moved to a new Kafka storage, whether the source is embedded stream or another external Kafka, because of how each solution stores data [R15]. The Helm chart guidance says the same for the stream tier: existing stream data is not migrated [R30]. What Pega offers instead is a controlled drain: the Stream Migration activity empties the queues so that no in-transit message is lost when the provider changes [R15].

Table 34. Treatment of each kind of Kafka content

Kafka content	Can it be copied?	Treatment
Queue processor messages (standard and dedicated)	No	Drain on the source; target starts with empty topics
Delayed queue processor items	No	Drain; items scheduled far in the future must be checked on the source before the drain (V-K-03)
Data flow stream partitions fed by the stream service	No	Drain; compare last processed IDs with topic end offsets [R30]
Stream data sets (application)	No	Producers stopped; consumers finish; target topics start empty
Consumer group offsets for Pega topics	Not needed	New topics, new offsets
Kafka data sets on a separate Kafka service	Not part of the stream service	Decision in Figure 22

11.3 Inventory

Build the inventory in staging and confirm it in production before each release. Record it in Appendix A.

Table 35. Kafka inventory

Inventory item	Where to find it	Recorded fields
Queue processors	Admin Studio > Queue processors	Name, type, status, Ready to process, Broken, node type
Job schedulers	Admin Studio > Job schedulers	Name, node type, schedule
Data flows reading streams	Data Flow landing page	Name, run ID, status, partitions
Kafka data sets and Kafka configuration instances	Records > Data Model > Data Set (type Kafka); Records > SysAdmin > Kafka	Name, brokers, topic, owner, consumer group
Topics on the target	Confluent CLI or console (command K-4)	Name, partitions, replication, max.message.bytes

11.4 Strategy

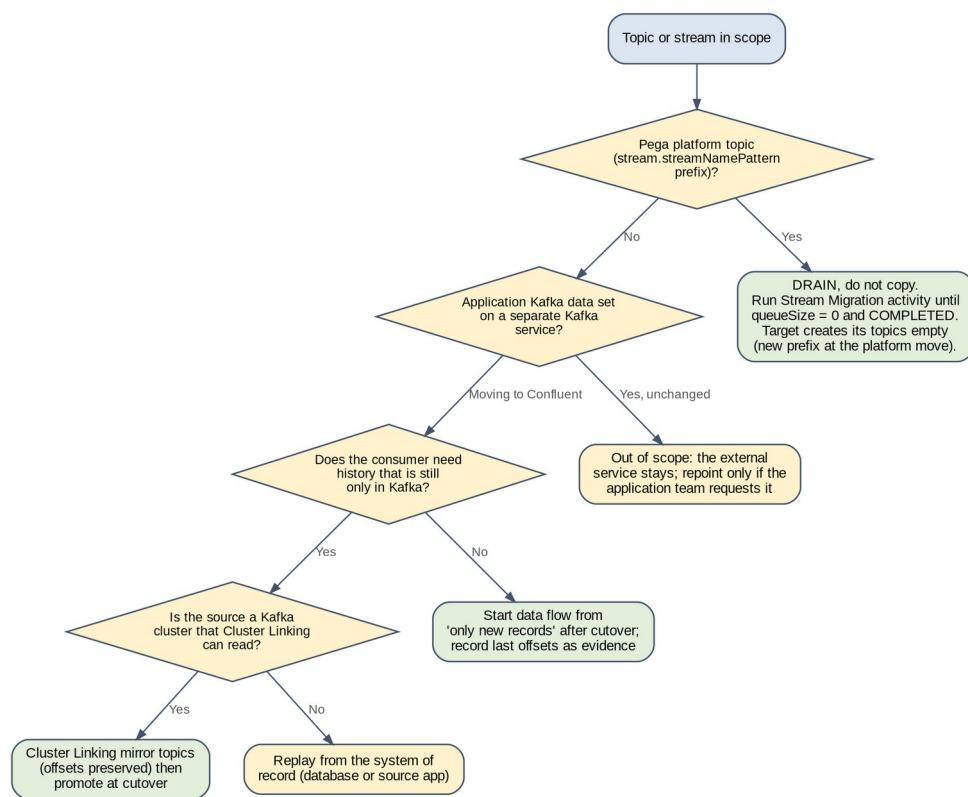


Figure 22. Decision: treatment of each topic or stream

Prepared for this implementation. Not a Pega-published diagram.

Table 36. Kafka strategy per release

Release	Kafka strategy	Why
Release 1	Drain the 8.8 stream with the Stream Migration activity, stop all nodes, configure Confluent, start. Topics are created under <code>pega- <env></code> on Confluent.	Pega documented procedure [R15]; no Pega software change at the same time [R30].
Release 2	Drain again with the activity, stop 8.8, update the copied database, start the bridge release with prefix <code>pega- <env>2</code> on the same Confluent cluster.	The new prefix gives the bridge empty topics and leaves the 8.8 topics untouched for rollback (AD-05).
Release 3	No drain. The zero-downtime update keeps the same prefix and topics.	Same Kafka cluster; Pega performs a rolling reboot [R28].

11.4.1 Application Kafka data sets

Kafka data sets belong to the application, not to the platform. Pega requires them to use a Kafka service separate from the stream service [R13]. For each one, the application owner chooses one of these options:

- **Keep the current Kafka service.** Only the network path from AKS changes. Confirm reachability and credentials in Release 2.
- **Move to a separate Confluent cluster, new records only.** The consuming data flow starts from new records after cutover; record the last offsets on the source as evidence.
- **Move to a separate Confluent cluster with history.** Use Confluent Cluster Linking: mirror topics copy messages byte for byte and keep partition and offset positions, and consumer offsets can be synchronized [R41]. Promote the mirror topics at cutover.
- **Replay from the system of record.** Use when the source cannot be linked and history is needed.

11.5 Validation matrix

Table 37. Kafka validation matrix

ID	When	Check	Method	Pass criterion	Evidence
V-K-01	Before	Target reachable from Pega pods	Commands K-1 to K-3 (Appendix D)	Private IPs; TLS chain valid; all brokers listed	Command output
V-K-02	Before	ACLs present	confluent kafka acl list	Four ACL entries per Table 20	ACL export
V-K-03	Before	Queues healthy	Admin Studio queue processors	All in scope RUNNING; Broken items resolved or accepted in writing	Screenshot and list
V-K-04	During	Drain complete	GET .../pzstream/migration	queueSize 0, timeToDrainMS 0, migrationStatus COMPLETED [R15]	JSON response with time
V-K-05	After start	Client compatibility	Pega log on start; produce and consume test queue item	No client-broker version errors; test item processed	Log extract
V-K-06	After start	Stream service healthy	Configure > Decisioning > Infrastructure > Services > Stream	Provider ExternalKafka; Status NORMAL [R15]	Screenshot
V-K-07	After start	Topic and ACL coverage	Command K-4; compare with ACL prefix	Every Pega-created topic under an allowed prefix or literal ACL	Topic list
V-K-08	After start	Message size limit	Command K-6 on each topic	max.message.bytes 5,000,000 or higher	Config output
V-K-09	After start	Queue processors running	Admin Studio	All expected queue processors RUNNING; Ready to process falling	Screenshot
V-K-10	After start	Partition budget	Command K-5	Total within limit with 30 % headroom	Count
V-K-11	After start	Application Kafka data sets	Run each data set test or data flow	Expected messages received at expected offsets	Data flow run report
V-K-12	After rollback window	Old topics removed	Command K-4 for the old prefix	No topics under the retired prefix	Topic list

11.6 Execution

Table 38. Kafka execution steps

Step	Action	Command or location	Expected result
K-E1	Confirm V-K-01 to V-K-03 pass.	Appendix D	All pass

Step	Action	Command or location	Expected result
K-E2	Stop producers: route users to a maintenance page, stop listeners and file or email services, pause producer data flows. Do not stop consumers.	Ingress maintenance rule; Admin Studio	No new items being queued
K-E3	Start the Stream Migration activity from inside a Pega pod.	<code>curl --request POST http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration [R15]</code>	HTTP success
K-E4	Poll the status every minute.	<code>curl --request GET http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration [R15]</code>	queueSize falls to 0; migrationStatus COMPLETED
K-E5	Stop all Pega nodes.	Scale tiers to 0 (AKS) or stop servers (VMs)	No nodes running
K-E6	Configure the target Kafka settings (Section 7.7).	Helm values or 8.8 stream settings	Configuration reviewed
K-E7	Start all nodes.	helm upgrade or server start	Pods Ready
K-E8	Run V-K-05 to V-K-11.	Section 11.5	All pass
K-E9	Only when switching between two external providers: restart the Kafka queues.	<code>curl --request DELETE http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration [R15]</code>	HTTP success; queues restarted
K-E10	Re-enable producers.	Remove maintenance rule; restart listeners	New items processed

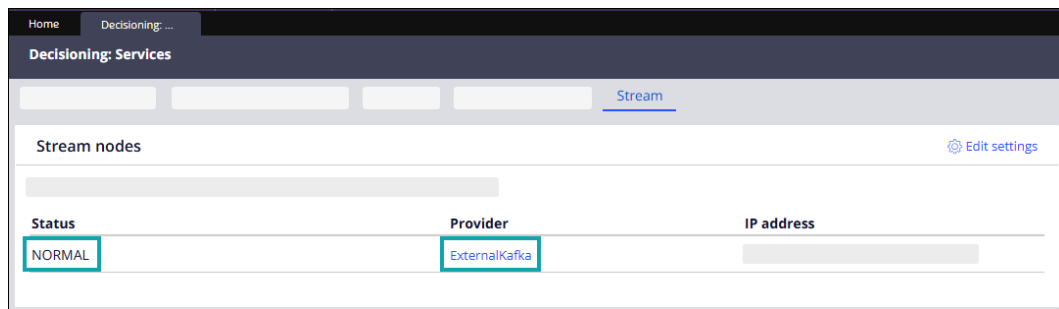


Figure 23. Stream landing page after the switch: Provider ExternalKafka, Status NORMAL

Source: Pega Documentation, "Switching Kafka providers while preserving Stream data" [R15]. © Pegasystems Inc. Reproduced with attribution.

Caution. The REST call uses localhost, so run it from inside a Pega node (for example `kubectl exec` into a batch pod). Run it once, from one node.

If the 8.8 source runs the deprecated stream tier on Kubernetes, the Helm chart describes an equivalent manual drain: set producing tiers to 0 replicas, wait until Ready to Process is 0 for every queue processor, and compare data flow partition last IDs with topic offsets from `GetOffsetShell` [R30]. Use it as a cross-check of the activity result.

11.6.1 Release 3 Kafka points

- Pega '26 replaces single-threaded queue processing with a partitioned model with isolated thread pools per processor [R3]. Expect different consumer group activity on Confluent after the update; compare throughput, not thread counts.
- If dynamic topic creation is allowed, Pega raises `SystemPulse_SystemPulseTopicName` to 6 partitions itself; otherwise do it before the update [R3].
- Confirm the Confluent cluster accepts the Kafka 4.0.0 client used by '26. Pega warns that a broker older than the client library can cause failures or slow client-broker communication [R3].

12 Search index migration: requirements, validation, strategy and execution

12.1 Requirements

Table 39. Search migration requirements

ID	Requirement	Source
R-S-01	After each release, every indexed class (Work, Data, Rules and custom indexes) must be searchable with results that match the database.	[R21]
R-S-02	The search cluster must be on the SRS compatibility matrix and configured with the two required cluster settings.	[R16]
R-S-03	Each Pega environment must use its own immutable customerDeploymentId.	[R17, R28]
R-S-04	Pega-to-SRS traffic must be authorized (OAuth) and encrypted (TLS or service mesh).	[R16]
R-S-05	The 8.8 indexes must remain usable until the Release 2 rollback window closes.	AD-06
R-S-06	The index build duration must fit inside the outage window, measured in rehearsal.	[R17]

12.2 What can and cannot be migrated

SRS builds its own indexes from the Pega database. After Pega connects to SRS, "SRS indexes all searchable data, which requires a downtime period" whose length depends on the data model, cluster resources, the number of queue processors and the amount of searchable data [R17]. Indexes from embedded search or the legacy plug-in are therefore not reused. Copying SRS indexes between search clusters with snapshots is also not used in this design, for two reasons: SRS owns index names and mappings, and Elasticsearch 8.x snapshots cannot be restored into OpenSearch, which can only restore snapshots from the versions before its fork, and supports moving later data only through reindexing tools [R43, R44].

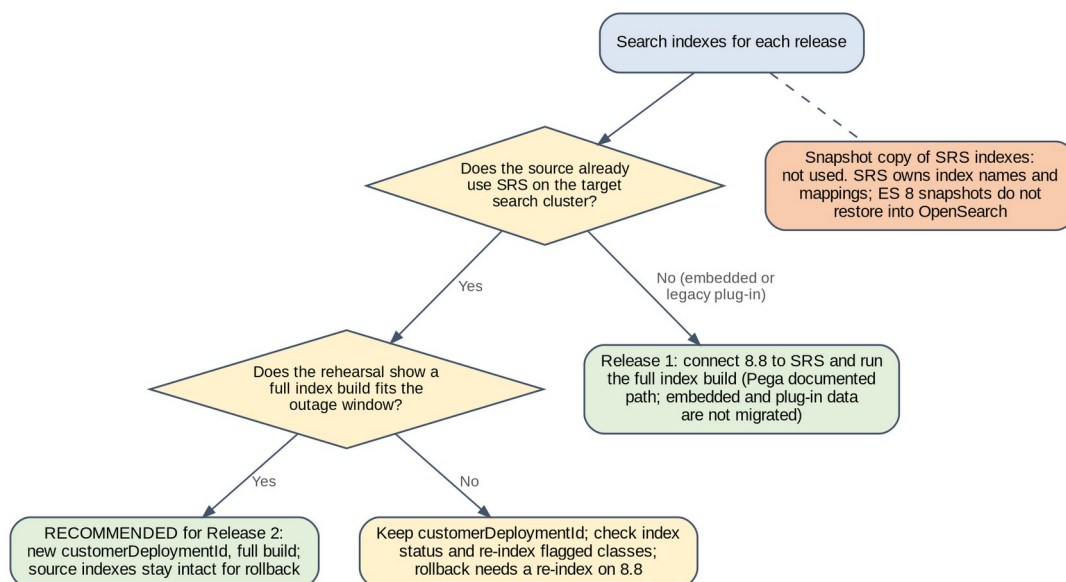


Figure 24. Decision: search index strategy

Prepared for this implementation. Not a Pega-published diagram.

12.3 Inventory and baseline

Table 40. Search inventory

Item	Where	Recorded fields
Indexed classes and status	Configure > System > Settings > Search, Indexing section, per tab [R21]	Class, index type, status, document count
Custom search indexes	Search landing page; custom search properties rules	Class, properties, owner
Database row counts for indexed classes	SQL count on each class table (DBA)	Count with timestamp
Index sizes on the search cluster	Command S-4	Index name, documents, size
Key business searches	Application team	Search text, expected result count

12.4 Strategy

Table 41. Search strategy per release

Release	Search strategy	Why
Release 1	Deploy SRS and the search cluster; connect 8.8 to SRS; run the full index build.	Pega documented path for moving to SRS [R17] .
Release 2	Set a new customerDeploymentId; the bridge release builds a full new set of indexes on the same search cluster.	Keeps the 8.8 indexes for rollback; SRS separates data by deployment ID [R16] .
Release 3	Keep the customerDeploymentId; check index status after the update; re-index only classes that report a problem.	Same SRS and indexes; zero-downtime update.

12.5 Validation matrix

Table 42. Search validation matrix

ID	When	Check	Method	Pass criterion	Evidence
V-S-01	Before	Cluster version and health	Commands S-1, S-2	Version on the matrix; status green	Output
V-S-02	Before	Required settings	Command S-3	auto_create_index false; destructive_requires_name false	Output
V-S-03	Before	SRS healthy	kubectl get pods -n srs-<env>; SRS logs	All replicas Ready; no storage connection errors	Output
V-S-04	Before	Pega reaches SRS	Command S-5 from a Pega pod	TLS handshake succeeds	Output
V-S-05	Before	customerDeploymentId	Helm values review	Set explicitly; new value for Release 2	Values diff
V-S-06	Before	OAuth token content	Decode a token from the IdP	Scope pega.search:full granted; guid equals customerDeploymentId	Decoded claims (no signature)
V-S-07	During	Build progress	Queue Processor landing page; system log [R19]	Progress advancing; no growing broken items	Screenshots
V-S-08	After	Index status	Search landing page [R21]	No class in an error or CONFLICTS FOUND status	Screenshot
V-S-09	After	Document counts	Search landing page counts against database counts	Equal for Work and Data classes, or differences explained	Comparison sheet
V-S-10	After	Business searches	Run the key searches	Expected results returned	Test record

ID	When	Check	Method	Pass criterion	Evidence
V-S-11	After	Incremental indexing	Create and update a case; search for it	Found within the agreed delay	Test record
V-S-12	After rollback window	Old indexes removed	Command S-4 with the old prefix	No indexes under the retired customerDeploymentId	Output

12.6 Execution

Table 43. Search execution steps

Step	Action	Where	Expected result
S-E1	Confirm V-S-01 to V-S-06 pass.	Appendix D	All pass
S-E2	Deploy or confirm SRS for the environment.	helm upgrade --install with backingservices values	SRS pods Ready
S-E3	Set pegasearch and global.customerDeploymentId in the Pega values (Section 8.7).	Pega Helm values	Values reviewed
S-E4	Start Pega. SRS begins indexing searchable data.	helm upgrade	Indexing queue items appear
S-E5	If an index needs a manual rebuild, open the Search landing page, select the index, click Re-index and choose the classes. The pxAccessSearchLP privilege is needed (PegaRULES:SysAdm4 includes it).	Configure > System > Settings > Search [R20]	Rebuild starts
S-E6	Monitor progress on the Queue Processor landing page and in the system log.	[R19]	Progress to completion
S-E7	If the build is slow and the database has headroom, tune DSS indexing/distributed/queue/maxrecords (records processed per invocation of the incremental indexer).	Dynamic System Settings [R19]	Higher throughput without database alerts
S-E8	Run V-S-07 to V-S-11.	Section 12.5	All pass

Note. Pega notes that reindexing individual classes leaves the state incomplete; a complete reindex is needed for the state to be available, and the system does not reindex child classes of a class you name [\[R20\]](#).

Rebuilds of Work classes have the largest performance effect [\[R21\]](#). Schedule manual rebuilds of large Work classes outside business hours after go-live.

13 Deployment runbook

This section builds each environment in the same order. Every step has a verification and an evidence item, so a reviewer can confirm the environment was built as designed. Values in angle brackets come from Appendix A.

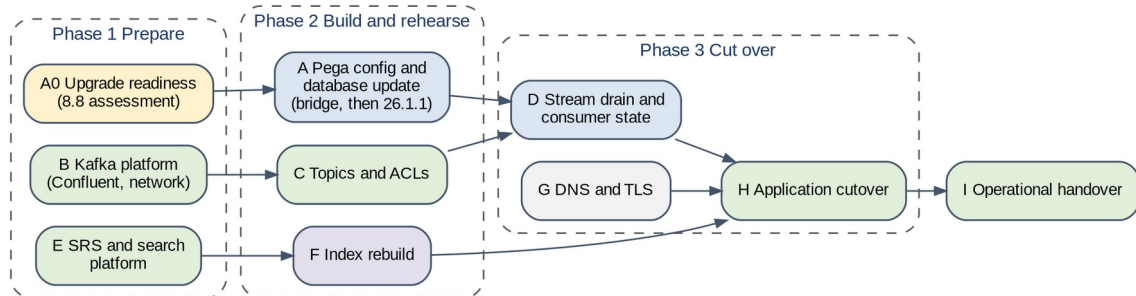


Figure 25. Workstreams and dependencies

Prepared for this implementation. Not a Pega-published diagram.

13.1 Build order

Table 44. Build order

Phase	Component	Depends on	Owner
1	Azure landing zone: VNet, subnets, firewall routes, private DNS zones, Key Vault, ACR	Customer landing zone	Platform team
2	AKS cluster, node pools, External Secrets Operator, monitoring agents	Phase 1	Platform team
3	Confluent network, cluster, private endpoints, DNS, service account, ACLs	Phase 1	Kafka engineer
4	Search cluster, private connectivity, cluster settings, SRS user	Phase 1	Search engineer
5	SRS deployment (backingservices chart)	Phases 2 and 4; OAuth client (OD-16)	Search engineer
6	Pega database and installer job; Pega tiers (pega chart)	Phases 2, 3 and 5	Pega LSA and platform team
7	Ingress, certificates, public DNS (switched at cutover)	Phase 6	Platform team

13.2 Azure foundation and AKS

Table 45. Azure foundation steps

ID	Action	Verification	Evidence
A-1	Create the spoke VNet and subnets (Section 6.3); peer with the hub; route 0.0.0.0/0 to the firewall.	Effective routes on the node NIC show the firewall next hop.	Route table export
A-2	Create private DNS zones and links (Section 6.3.2).	Resolution test from a VM in the spoke.	nslookup output
A-3	Create Key Vault (RBAC model, private endpoint, public access disabled) and ACR (private endpoint).	Access from the spoke works; from the internet is refused.	Test output

ID	Action	Verification	Evidence
A-4	Create the private AKS cluster with Azure CNI, workload identity and OIDC issuer enabled; system, pega and srs node pools across three zones.	kubectl get nodes -L topology.kubernetes.io/zone shows three zones.	Command output
A-5	Grant the kubelet identity AcrPull on ACR.	A test pod pulls a Pega image.	Pod event log
A-6	Install External Secrets Operator; create the workload identity federated credential and grant Key Vault Secrets User.	An ExternalSecret reaches status SecretSynced.	kubectl describe output
A-7	Install the ingress controller for Application Gateway; create the WAF policy.	Health probe of a test service is healthy.	Portal screenshot
A-8	Create namespaces pega-<env>, srs-<env>, platform-ops; apply network policies (deny by default; allow pega to srs, srs to search endpoint, pods to DNS).	Policy test pod cannot reach a denied target.	Test output

13.3 Confluent Cloud

Table 46. Confluent Cloud steps

ID	Action	Verification	Evidence
C-1	Create the Confluent environment and network in the AKS region; create the cluster (type per OD-05).	Cluster status Up in the console.	Cluster ID
C-2	Complete N-1 to N-6 (Section 7.4).	V-K-01 passes.	Command output
C-3	Create a service account per Pega environment and one API key for it; store the JAAS string in Key Vault.	Key Vault secret version exists; no key in any file or ticket.	Secret metadata
C-4	Create the ACLs (Section 7.5.2).	V-K-02 passes.	ACL export
C-5	Agree the partition budget (Section 7.6) and confirm the cluster capacity.	Budget recorded in Appendix A.	Calculation
C-6	Enable audit logs and metrics export (Section 18).	Metrics visible in the monitoring workspace.	Screenshot

13.4 Search cluster and SRS

Table 47. Search and SRS steps

ID	Action	Verification	Evidence
S-1	Provision the search cluster per OD-04 with three cluster-manager and three or more data nodes across zones.	V-S-01 passes.	Output
S-2	Enable private connectivity and DNS for the search endpoint.	Resolution and TLS test from a pod in srs-<env>.	Output
S-3	Create the SRS user. Grant the manage cluster privilege, or apply the settings in Section 8.5 manually.	V-S-02 passes.	Output
S-4	Register the OAuth client for Pega (private_key_jwt preferred) and record the token endpoint and public key URL.	V-S-06 passes.	Decoded claims
S-5	Create the Key Vault secrets and ExternalSecrets for srs-search-credentials, srs-runtime-certs and pega-srs-oauth.	Kubernetes secrets exist.	kubectl output
S-6	Deploy SRS: helm upgrade --install srs-<env> pega/backingservices -n srs-<env> -f backingservices-<env>.yaml.	V-S-03 and V-S-04 pass.	Output

13.5 Pega

Table 48. Pega deployment steps

ID	Action	Verification	Evidence
G-1	Prepare values-<env>.yaml from Appendix B: provider aks, images, jdbc with external secret, stream, pegasearch, customerDeploymentId, tiers, ingress, hpa, pdb.	Peer review against Appendix A.	Reviewed values file
G-2	Add the Pega Helm repository: helm repo add pega https://pegasystems.github.io/pega-helm-charts ; pin the chart version.	helm search repo pega shows the pinned version.	Output
G-3	Run the installer action (install for a new environment, upgrade for a migrated database).	Installer job completes successfully.	Job log
G-4	Deploy the tiers (global.actions.execute deploy).	All pods Ready; P-1.	Output
G-5	Log in as administrator; check the Stream landing page.	V-K-06 passes.	Screenshot
G-6	Check queue processors and search status.	V-K-09 and V-S-08 pass.	Screenshots
G-7	Run the smoke test pack (Section 16).	All smoke tests pass.	Test report

13.6 Promotion across environments

- Keep values files, ExternalSecret manifests, Confluent ACL scripts and search settings in Git. Every change goes through review.
- Build environments in order: development, test, staging, production. Do not change a setting in production that was not first applied in staging.
- Pin the Pega Helm chart version and image digests per release. Record them in Appendix A.
- Each environment has its own topic prefix, service account, customerDeploymentId and OAuth client.

14 Cutover runbook

This section describes the Release 2 cutover in detail, because it has the longest outage. Release 1 follows the sequence in Section 10.5 and Release 3 the sequence in Section 10.7; both use the same Go/No-Go structure.

14.1 Go/No-Go criteria

Table 49. Go/No-Go gates

Gate	When	Criteria	Decision owner
Go/No-Go 1	Seven days before	Two successful rehearsals on staging with production-scale data; durations fit the window with at least 25 % contingency; all V-K and V-S checks passed in rehearsal; rollback rehearsed; open defects accepted	Business owner with platform lead
Go/No-Go 2	After smoke tests, before DNS switch	V-K-06, V-K-09, V-S-08, V-S-09 pass; smoke tests pass; Hazelcast checks H-6 and H-7 pass	Cutover manager
Go/No-Go 3	30 minutes after DNS switch	No priority 1 or 2 incidents; error rate and response times within agreed limits; queue backlog falling	Cutover manager with business owner

14.2 Release 2 cutover sequence

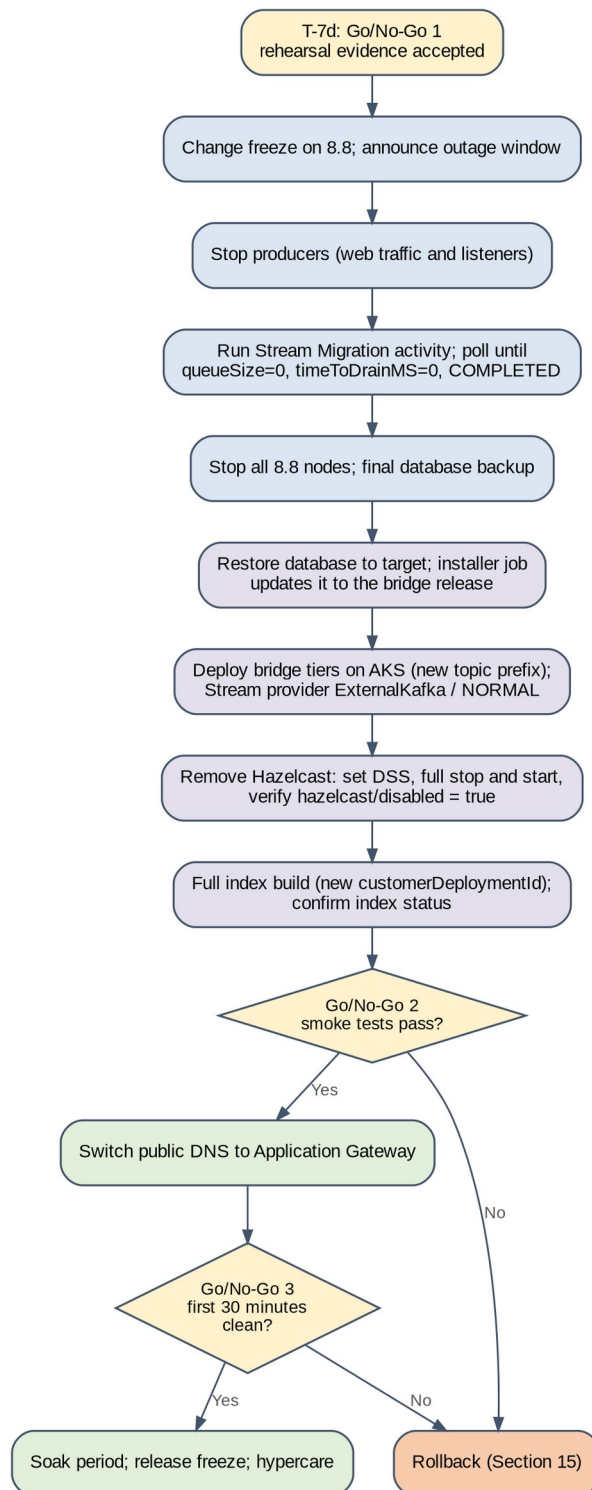


Figure 26. Release 2 cutover sequence

Prepared for this implementation. Not a Pega-published diagram.

Table 50. Release 2 cutover steps

Step	Action	Owner	Duration source	Evidence
T-1	Confirm Go/No-Go 1 record; confirm backups and support contacts (Pega, Confluent, search provider, Azure).	Cutover manager	Fixed	Signed record
T-2	Start change freeze on 8.8; publish the outage notice.	Business owner	Fixed	Notice
T-3	Stop producers (K-E2).	Pega LSA	Rehearsal	Screenshot
T-4	Drain with the Stream Migration activity (K-E3, K-E4).	Pega LSA	Rehearsal	JSON responses (V-K-04)
T-5	Stop all 8.8 nodes; take the final database backup.	Platform team, DBA	Rehearsal	Backup ID
T-6	Restore the backup to the Azure database.	DBA	Rehearsal	Restore log
T-7	Run the bridge installer job (upgrade, in-place).	Platform team	Rehearsal	Job log
T-8	Deploy the bridge tiers with the new prefix and customerDeploymentId.	Platform team	Rehearsal	Pods Ready
T-9	Remove Hazelcast (H-1 to H-7).	Pega LSA	Rehearsal	API responses
T-10	Full index build; V-S-07 to V-S-09.	Pega LSA	Rehearsal	Screenshots, comparison sheet
T-11	Smoke tests; Go/No-Go 2.	Test lead	Rehearsal	Test report, gate record
T-12	Switch public DNS to Application Gateway; re-enable producers.	Platform team	DNS TTL	DNS change record
T-13	Go/No-Go 3; start hypercare.	Cutover manager	30 minutes	Gate record

Important. Lower the TTL of the public DNS record to 300 seconds at least 48 hours before cutover so that a DNS switch or revert takes effect quickly. Restore the normal TTL after hypercare.

14.3 Communication

Table 51. Communication plan

When	Audience	Message	Channel
14 days before	Business users	Outage window and expected effect	Email and intranet
1 day before	Support teams and vendors	Contacts, bridge call details, escalation path	Email
Each gate	Steering group	Gate result and next step	Bridge call and chat
Go-live	Business users	Service available; how to report issues	Email

15 Rollback

15.1 Principles

- Each release keeps the previous state intact until its rollback window closes: 8.8 topics and indexes in Releases 1 and 2, the 8.8 database and servers in Release 2, the pre-update backup and old rules schema in Release 3.
- Rollback is simplest before users work on the new state. After that, data entered on the new state must be reconciled or re-keyed, so the business owner decides between rollback and fixing forward.
- The rollback window and point of no return for each release are agreed at Go/No-Go 1 (OD-10).

15.2 Release 1 rollback

Table 52. Release 1 rollback

Situation	Action	Data effect
Before traffic is re-opened	Stop nodes; restore the previous stream and search configuration; start nodes.	None: queues were drained and no new work was created.
After traffic is re-opened	Prefer fixing forward. A move from Confluent back to the 8.8 embedded stream is not a documented Pega path; if the business still decides to revert, drain with the Stream Migration activity first and accept the risk in writing.	Messages produced to Confluent must be drained first; searches use the old indexes, which miss changes made after the switch until reindexed.

15.3 Release 2 rollback

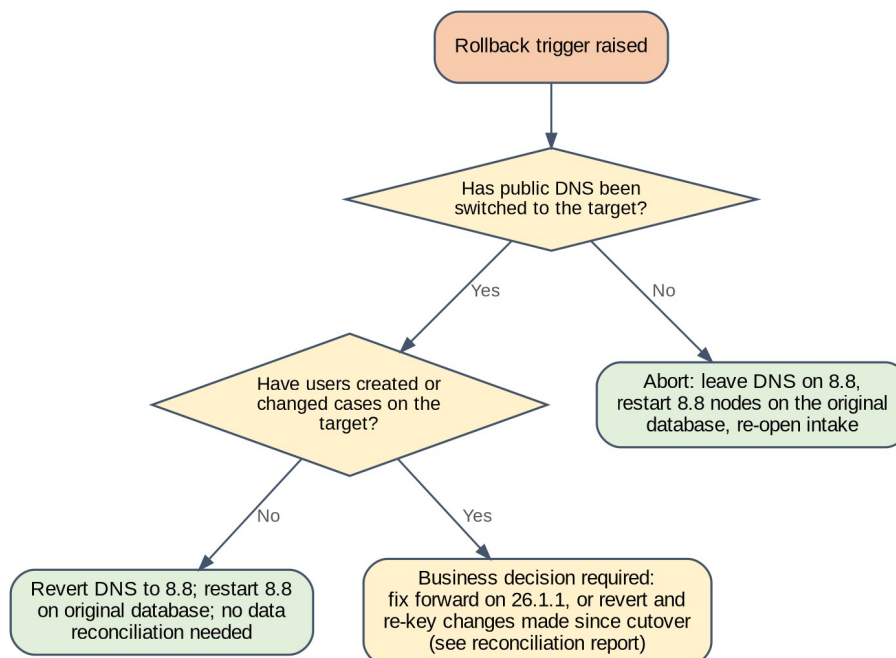


Figure 27. Decision: Release 2 rollback

Prepared for this implementation. Not a Pega-published diagram.

Table 53. Release 2 rollback steps

Step	Action
RB-1	Declare rollback and record the reason and time.
RB-2	Stop producers on the target. If users worked on the target, export the list of cases created or changed since cutover (reconciliation report).
RB-3	Revert public DNS to the 8.8 entry point.
RB-4	Start the 8.8 nodes on the original database. Their topics (old prefix) and indexes (old customerDeploymentId) were not touched.
RB-5	Check the 8.8 Stream landing page (Provider ExternalKafka, Status NORMAL) and search status.
RB-6	Re-open 8.8 to users; scale the target to zero but keep it for analysis.

15.4 Release 3 rollback

The zero-downtime update upgrades the data schema in place after the rules move to the new rules schema [R28]. Rolling back therefore needs the pre-update database backup. Take a full backup immediately before U-7, keep the old rules schema until acceptance, and redeploy the bridge images against the restored database if a rollback is declared. Work done on 26.1.1 after the update is lost on rollback unless re-keyed, so the business owner makes this decision.

16 Testing and acceptance

16.1 Test stages

Table 54. Test stages

Stage	Purpose	Examples	Environment
Connectivity	Prove every network and security hop	V-K-01 to V-K-03, V-S-01 to V-S-06	Every environment
Component	Each service works on its own	Produce and consume with kcat; SRS health; index create through SRS	Development
Integration	Pega uses Kafka and search correctly	Queue processor test item; job scheduler run; case search; report on indexed data	Test
Application regression	Business functions unchanged	PegaUnit test suites; automated UI and API tests; manual scripts	Test, staging
Performance	Throughput and response time at peak load	Peak user load plus queue processor load; index build throughput	Staging
Resilience	Behavior when a component fails	See Section 16.4	Staging
Security	Controls work as designed	No public endpoints; secret scanning of values files; token claim checks; TLS versions	Test, staging
Migration rehearsal	Measure and prove the cutover	Full Release 1, 2 or 3 run with timing	Staging
User acceptance	Business sign-off	Business scenarios including search	Staging
Operational acceptance	Run the service	Alerts fire and reach on-call; runbooks used by operations staff	Staging

16.2 Rehearsals

Plan at least two rehearsals for each release: a technical rehearsal to prove the steps, and a dress rehearsal run by the people who will do the production cutover, against production-scale data (OD-13). Record these measurements, because they set the outage window:

Table 55. Rehearsal measurements

Measurement	Steps	Used for
Drain duration	K-E3 to K-E4	Window; producer stop time
Database backup, transfer and restore	T-5 to T-6	Window; database copy method (OD-17)
Installer job duration	T-7, U-7	Window
Hazelcast removal	H-1 to H-7	Window
Full index build	S-E4 to S-E6	Window; search sizing
Smoke test duration	T-11	Window
Partition count after start	K-5	Confluent capacity

16.3 Pega unit and integration tests

- Run the application's PegaUnit test suites after each release in test and staging; record pass rates against the 8.8 baseline.

- Add a platform test case that queues an item to a test queue processor and asserts it is processed; run it after every deployment.
- Add a search test case that creates a case, waits for indexing, and searches for it.
- For each Kafka data set, run its data flow in test and compare record counts with the source.

16.4 Resilience tests

Table 56. Resilience tests

Test	Method	Expected behavior
Batch pod loss	Delete one batch pod during queue processing	Pod is recreated; queue processing continues; no items lost
Web pod loss	Delete one web pod under load	Users on other pods unaffected; affected users re-authenticate
Zone loss (simulated)	Cordon and drain all nodes in one zone	Pods reschedule in other zones within PDB limits
SRS pod loss	Delete one SRS pod	Searches continue on remaining replicas
Search node restart	Restart one data node (provider tooling)	Cluster yellow then green; searches continue
Secret rotation	Rotate the Confluent API key (Section 9.1)	No processing gap beyond the rolling restart

16.5 Acceptance criteria

Table 57. Acceptance criteria

ID	Criterion
AC-1	All validation checks in Sections 11.5 and 12.5 pass in production, with evidence filed.
AC-2	Application regression pass rate equal to or better than the 8.8 baseline; no open priority 1 or 2 defects.
AC-3	Performance results within the targets agreed by the business owner.
AC-4	Resilience tests pass in staging.
AC-5	Monitoring and alerts in Section 18 are live and tested.
AC-6	Operations staff have run the routine procedures in Section 18.3 in staging.

17 Troubleshooting

17.1 Method

Most failures between Pega and Kafka or search are connectivity failures. Work through the chain in Figure 28 in order; a failure at one step makes every later step fail, so the first failing step is the one to fix. Commands are in Appendix D.

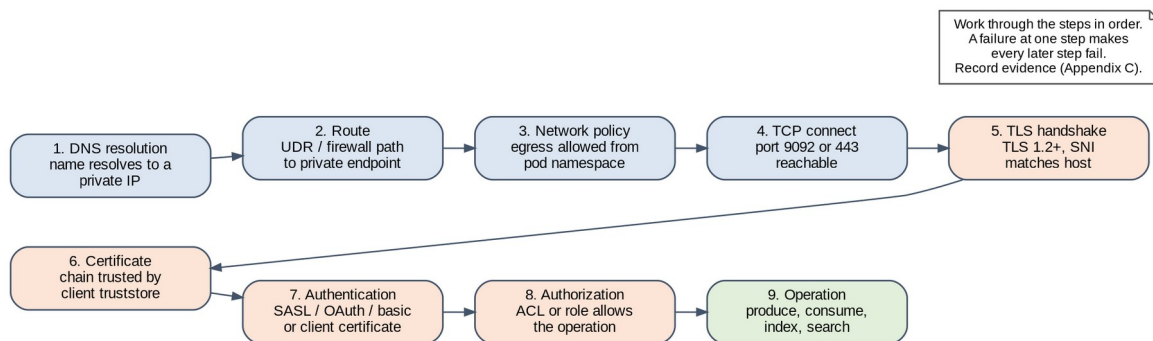


Figure 28. Connectivity chain for Kafka and search
Prepared for this implementation. Not a Pega-published diagram.

17.2 Kafka

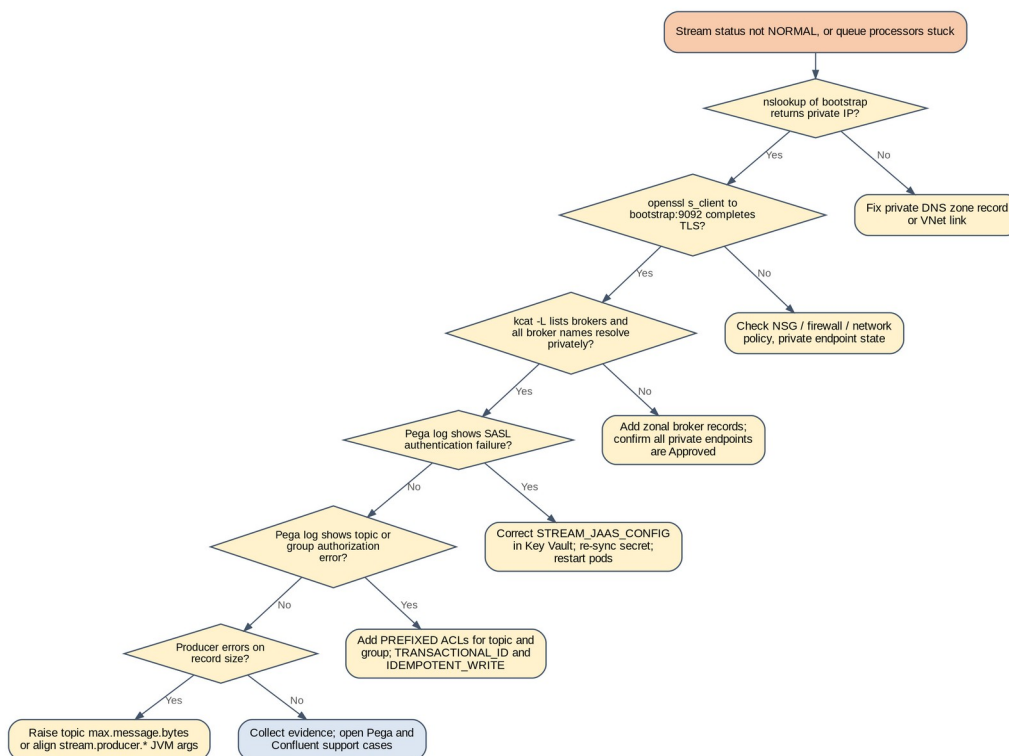
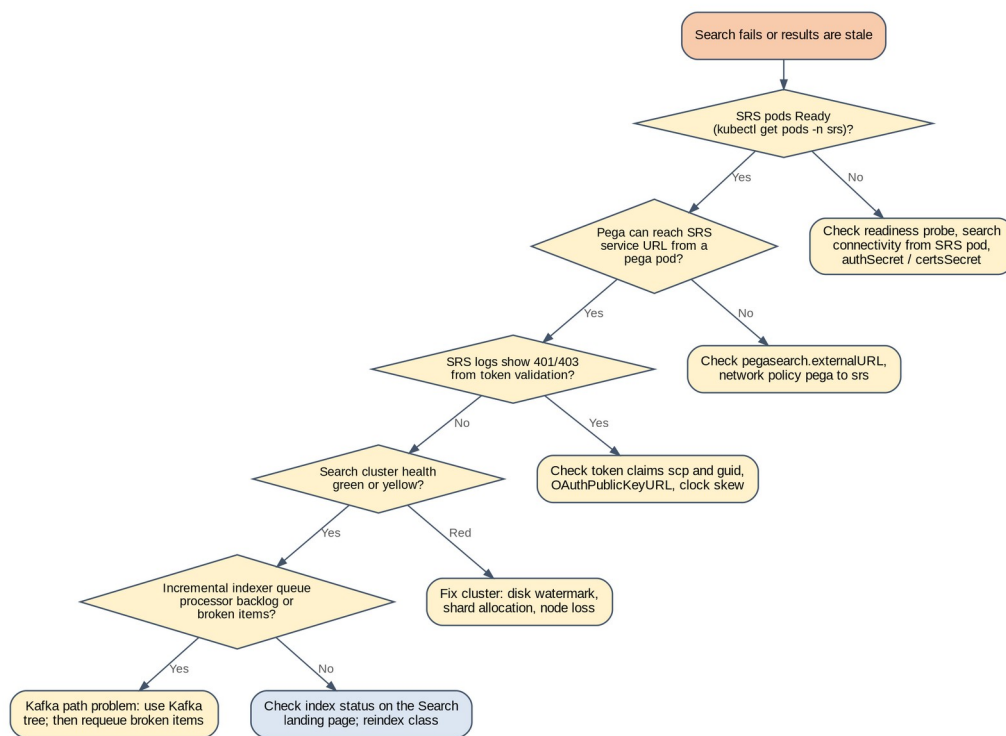


Figure 29. Troubleshooting tree: stream service
Prepared for this implementation. Not a Pega-published diagram.

Table 58. Kafka symptoms

Symptom	Likely cause	Check	Fix
Stream landing page status not NORMAL after start	Bootstrap unreachable or authentication failure	K-1 to K-3; Pega log for SASL errors	Fix DNS or network; correct the JAAS secret and restart
Connects to bootstrap, then broker timeouts	Zonal broker DNS records missing	Resolve each broker name from K-3	Add the zonal records Confluent lists [R38]
TopicAuthorizationException or GroupAuthorizationException	ACL prefix does not match streamNamePattern	Compare V-K-07 output with ACLs	Correct the prefix or add the ACL
Transactional or idempotent producer errors	Missing TRANSACTIONAL_ID or IDEMPOTENT_WRITE ACL; Freight cluster	ACL list; cluster type	Add ACLs; use Enterprise or Dedicated [R37]
RecordTooLargeException	Topic max.message.bytes below the message size	K-6	Raise to 5,000,000; align JVM arguments if larger [R13]
Queue processors stuck with Ready to process growing	Batch tier down, or consumers failing	Admin Studio; batch pod logs	Restore batch pods; fix the failing processor
Slow client-broker communication after Release 3	Client and broker compatibility	Pega log; Confluent support	Confirm the Confluent version supports the 4.0.0 client [R3]
Stream Migration activity never reaches COMPLETED	Items in STOPPED, FAILED or broken state; producers still active	Admin Studio; data flow status	Stop producers; restart or resolve stuck processors; poll again [R15]

17.3 Search

**Figure 30. Troubleshooting tree: search**

Prepared for this implementation. Not a Pega-published diagram.

Table 59. Search symptoms

Symptom	Likely cause	Check	Fix
SRS pods not Ready	Search cluster unreachable or wrong credentials	SRS logs; S-1 from an SRS pod	Fix network or authSecret; restart SRS
Pega search returns errors with 401 or 403 in SRS logs	Token missing scope or guid; wrong public key URL	Decode a token (V-S-06)	Fix the IdP claims (OD-16) or OAuthPublicKeyURL
Index creation fails	Index auto-creation or permissions	S-3; SRS user privileges	Apply Section 8.5 settings; grant privileges
Index deletion fails on Elasticsearch 8	destructive_requires_name true	S-3	Set it to false [R16]
Results missing for recent changes	Incremental indexing backlog	Queue processor landing page	Clear the backlog; check batch capacity
Class status CONFLICTS FOUND	Mapping conflicts in indexed properties	Search landing page [R21]	Analyze and fix the conflicting properties; reindex the class
Cluster status red	Lost data node or disk watermark	S-2; provider console	Restore capacity; follow provider guidance

17.4 Upgrade and Hazelcast removal

Table 60. Upgrade symptoms

Symptom	Likely cause	Fix
primaryKeyUtility reports insufficient database space	The utility copies each table missing a primary key	Increase database storage and rerun [R12]
ClassNotFoundException for javax.* classes after Release 3	Custom JAR not rebuilt for Jakarta EE	Import the updated JAR into a higher CodeSet version and restart [R11]
checkRemoteExecutionConnectivity does not return allNodesReachable	Nodes cannot exchange messages through Kafka	Check the stream service and topic ACLs; rerun [R7]
Installer job fails	Database connectivity, permissions or schema names	Read the job log (P-5); correct values; rerun

17.5 Support cases

Pega Global Client Support handles issues in externalized services with the same model as for databases and application servers: basic troubleshooting to determine the cause, then a handover to the third-party provider if the issue is not in Pega software [\[R24\]](#). Open cases with this evidence: Pega logs from all tiers for the period, Stream landing page screenshot, queue processor status, SRS logs, the relevant Appendix D outputs, and the Helm values with secrets removed.

18 Observability and operations

18.1 Monitoring design

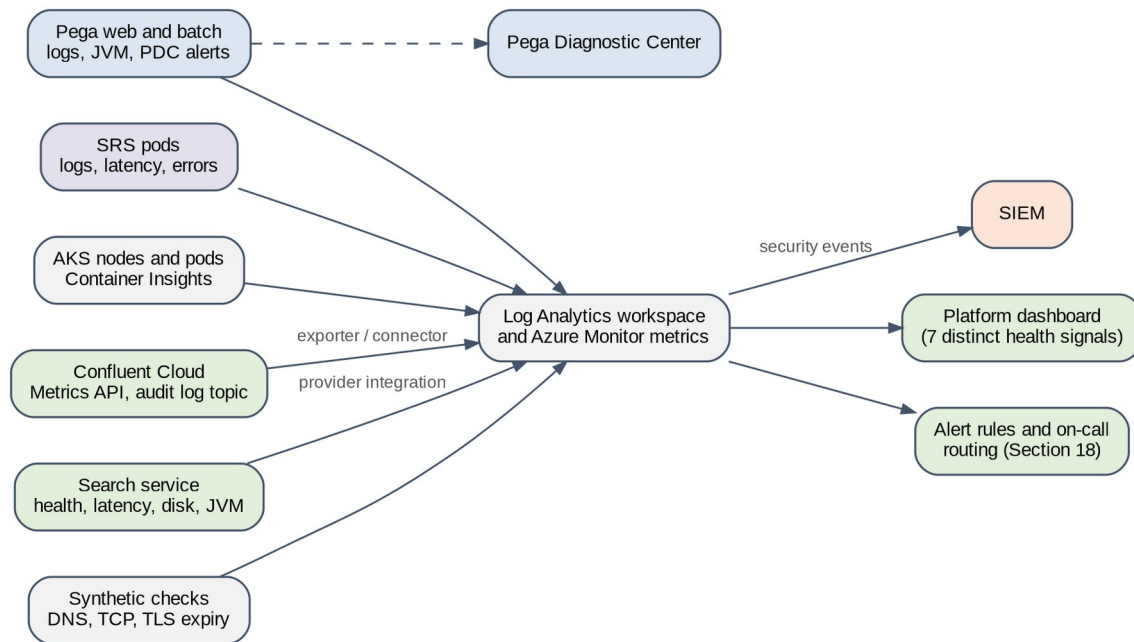


Figure 31. Telemetry flows

Prepared for this implementation. Not a Pega-published diagram.

18.2 Signals and alerts

Table 61. Signals and alerts

Signal	Source	Alert when	First action
Stream service status	Pega (Stream landing page, PDC)	Not NORMAL	Section 17.2
Queue processor backlog	Pega Admin Studio, PDC	Ready to process grows for longer than the agreed period	Check batch tier and consumer errors
Broken queue items	Pega	Above zero for business-critical processors	Investigate and requeue
Confluent consumer lag and request errors	Confluent Metrics API	Lag above baseline; authentication or authorization errors	Section 17.2
Confluent partition count	Confluent Metrics API	Above 70 % of the cluster limit	Plan capacity (Section 7.6)
SRS errors and latency	SRS logs and metrics	Error rate or latency above baseline	Section 17.3
Search cluster health and disk	Provider metrics	Status yellow for longer than the agreed period, red at once; disk above watermark warning	Section 17.3
Pod restarts and readiness	Container Insights	Restart loops; pods not Ready	kubectl describe; logs
Certificate expiry	Synthetic checks	Less than 30 days to expiry	Renew (Section 9.2)
Private endpoint DNS	Synthetic checks	Resolution returns a public IP or fails	Section 6.3.2

18.3 Routine operations

Table 62. Routine operations

Task	Frequency	Owner	Procedure
Check new Pega topics for max.message.bytes	After each deployment	Kafka engineer	K-4, K-6
Rotate Confluent API keys	Per security policy	Kafka engineer	Section 9.1
Rotate search credentials	Quarterly	Search engineer	Update Key Vault; restart SRS
Renew certificates	Before expiry	Platform team	Section 9.2
Review partition and storage growth	Monthly	Kafka and search engineers	K-5; S-4
Apply Pega patches	Per Pega maintenance program	Pega LSA	Zero-downtime patch [R28]
Update SRS	With Pega updates or security fixes	Search engineer	Section 8.6
Upgrade AKS	Per AKS support calendar	Platform team	Node pool surge upgrade with PDBs
Delete retired topics and indexes	After each rollback window	Kafka and search engineers	V-K-12, V-S-12

19 Resilience and disaster recovery

Within the primary region, every layer runs across three availability zones: AKS node pools, the Confluent cluster, the search cluster and the database. The disaster recovery design for a region loss is an open decision (OD-12). The points below apply whichever option is chosen.

Table 63. Regional disaster recovery points

Layer	Regional loss behavior	Design point
Pega tiers	Redeploy in the DR region from Git and ACR (geo-replicated)	Keep DR values files current; test yearly
Database	Recover from geo-replica or geo-backup	RPO and RTO set by the database option
Kafka	Stream data in flight in the lost region is not recoverable through Pega; the DR cluster starts with empty Pega topics, the same principle as in migration [R15]	Pre-create the DR Confluent cluster, network and ACLs; accept loss of in-flight queue items, or keep them recoverable from the database
Search	Rebuild indexes in DR from the database	Plan the rebuild time in the RTO; or use the provider's cross-region replication if offered
Secrets	Key Vault in DR region with replicated secrets	Separate Confluent service account and OAuth client for DR

20 Roles and responsibilities

R = responsible, A = accountable, C = consulted, I = informed.

Table 64. RACI

Activity	Business owner	Enterprise architect	Security	Platform team	Pega LSA	Kafka engineer	Search engineer	Test lead
Approve design and open decisions	A	R	C	C	C	C	C	I
Azure foundation and AKS	I	C	C	A/R	I	I	I	I
Confluent Cloud	I	C	C	C	C	A/R	I	I
Search cluster and SRS	I	C	C	C	C	I	A/R	I
Pega configuration and update	I	C	I	R	A/R	C	C	I
Kafka topic migration	I	I	I	C	A/R	R	I	C
Search index migration	I	I	I	C	A/R	I	R	C
Testing and rehearsals	C	I	C	R	R	R	R	A
Cutover and rollback decisions	A	C	I	R	R	R	R	C
Operations after handover	I	I	C	A/R	R	R	R	I

21 Risks

Table 65. Risk register (L likelihood, I impact: H high, M medium, L low)

ID	Risk	L	I	Mitigation	Owner
RK-01	The bridge-to-26.1.1 path is not confirmed for the chosen bridge patch	M	H	OD-02: confirm with Pega Support before Release 2 design freeze	Pega LSA
RK-02	8.8 servers cannot reach Confluent and SRS privately for Release 1	M	H	Network design early; fallback is the combined option after Pega Support confirmation	Platform team
RK-03	Index build longer than the outage window	M	H	Measure in rehearsal; scale search data nodes and SRS for the build; tune maxrecords DSS	Search engineer
RK-04	Drain does not complete because of stuck or broken items	M	M	Resolve broken items before cutover; producer stop checklist; time-box with a Go/No-Go decision	Pega LSA
RK-05	Identity provider cannot issue the claims SRS checks	M	H	OD-16: prove in the first environment; choose another provider if needed	Security
RK-06	Partition limit reached on Confluent Enterprise	L	M	Measured budget; Dedicated if needed	Kafka engineer
RK-07	Messages larger than the topic limit after Pega creates new topics	M	M	Post-deployment topic check (V-K-08); alert	Kafka engineer
RK-08	Custom JARs fail on Jakarta EE	M	H	Upgrade Tools report early; rebuild and test in Release 3 rehearsal	Pega LSA
RK-09	Tables without primary keys block the update	M	H	Dry run in preparation; fix before Release 3	DBA
RK-10	Behavior change in queue processing on '26	M	M	Custom queue processor tests (P-6); performance test	Pega LSA
RK-11	Rehearsal data not representative	M	H	OD-13: production-scale masked or synthetic data	Test lead
RK-12	Vendor documentation changes between rehearsal and release	M	M	Recheck Section 3 facts before each release	Enterprise architect

22 Open decisions

The draft runbook used OD-01 to OD-15. This register replaces that list for Kafka, search and the upgrade path.

Table 66. Open decisions

ID	Decision	Options	Recommendation	Owner	Needed by
OD-01	Source estate facts	Collect per Section 4.1	Complete the assessment	Pega LSA	Design start
OD-02	Bridge release patch	Latest '24.1 (24.1.4+); latest '24.2 (24.2.3+)	Latest '24.1, confirmed with Pega Support	Pega LSA	Release 2 design
OD-03	Delivery option	Three releases; combined Release 1 and 2	Three releases	Business owner	Plan approval
OD-04	Search provider	Elastic Cloud on Azure; managed OpenSearch provider on Azure; self-managed OpenSearch	Managed service that passes the Section 8.3 checks	Enterprise architect	Release 1 design
OD-05	Confluent cluster type	Enterprise; Dedicated	Enterprise unless partition budget or networking needs Dedicated	Kafka engineer	Release 1 design
OD-06	Kafka authentication	SASL/PLAIN API key; SASL/OAUTHBEARER	PLAIN; review OAUTHBEARER after go-live	Security	Release 1 design
OD-07	Topic creation	Pega creates (CREATE ACL); pre-created topics	Pega creates	Security	Release 1 design
OD-08	Application Kafka data sets	Keep; new only; Cluster Linking; replay	Per data set (Section 11.4.1)	Application owner	Release 2 design
OD-09	Outage windows	Per release	From rehearsal plus 25 %	Business owner	Go/No-Go 1
OD-10	Rollback windows and points of no return	Per release	Agree at Go/No-Go 1	Business owner	Go/No-Go 1
OD-11	Index strategy in Release 2	New customerDeploymentId with full build; reuse	New ID with full build	Pega LSA	Release 2 rehearsal
OD-12	Disaster recovery	Warm standby; rebuild in DR	Business RPO and RTO first	Enterprise architect	Release 2 design
OD-13	Rehearsal data	Masked production; synthetic at scale	Production-scale, masked	Business owner	First rehearsal
OD-14	Monitoring integration	Azure Monitor; existing tools	Azure Monitor with SIEM export	Platform team	Release 1
OD-15	Third-party support contracts	Confluent and search provider support levels	Production support for both	Business owner	Release 1
OD-16	Identity provider for Pega to SRS	Entra ID with custom claims; another OAuth provider	Provider that issues scope and guid claims	Security	First environment
OD-17	Database copy method	Native backup and restore; replication tool	Measured in rehearsal	DBA	Release 2 rehearsal

23 Source reconciliation register

These are the places where sources disagree, or where an earlier draft or review said something the evidence does not support. Each entry states the action taken in this document.

Table 67. Reconciliation register

ID	Topic	What the sources say	Action
RC-01	SRS and search versions	'26 page: SRS 1.44.3 or later; Elasticsearch up to 8.19.11, best practice 8.18.3; OpenSearch best practice 2.15 [R16] . Helm README: SRS 1.49.2, Elasticsearch up to 8.19.22, OpenSearch 2.19 [R29] .	Choose search versions listed on the '26 page; use the SRS image the README lists for the chart in use; recheck before each release.
RC-02	Kafka SASL mechanisms	Pega page lists JAAS, OAUTHBEARER, PLAIN and SCRAM [R13] ; Helm saslMechanism lists PLAIN, SCRAM-SHA-256, SCRAM-SHA-512 [R28] .	Use PLAIN (AD-08). OAUTHBEARER only after a successful rehearsal.
RC-03	Kafka version	KafkaClusterRequirement.md says Apache Kafka 3.4.0 or earlier [R31] ; '26 ships client 4.0.0 and supports Kafka server 4.0.0 [R3] .	Follow the '26 documentation.
RC-04	Broker settings on Confluent Cloud	Pega lists replica.fetch.* and unclean.leader.election.enable [R13] ; these are not editable on Confluent Cloud [R40] .	Request written confirmation from Confluent; set topic max.message.bytes (V-K-08).
RC-05	SystemPulse partitions	Hazelcast removal prerequisites: one partition [R6] ; '26 prerequisites: six partitions [R3] .	One at the bridge release if pre-creating topics; six for '26.
RC-06	Extra partitions for Hazelcast removal	About 100 per environment; 30 from 25.1.2 or 25.1.1 with HFIX-C4655 [R6] .	Plan 100 for the '24.1 bridge.
RC-07	Confluent Private Link on Azure	The Azure Private Link page describes Dedicated clusters [R38] ; Enterprise uses Private Link for serverless products.	Follow the Confluent page that matches the cluster type chosen in OD-05.
RC-08	Hazelcast Clustering Service	An earlier review of the draft recommended deploying the Hazelcast Clustering Service. Pega removed Hazelcast in '25, including the Clustering Service [R5, R9] .	Withdrawn. Hazelcast is removed in Release 2 and not used on 26.1.1.
RC-09	Direct update from 8.8	The draft assumed one update from 8.8 to 26.1.1. Hazelcast removal tooling needs 23.1.4, 24.1.3, 24.2.2 or later [R9] .	Replaced by the three-release plan (Section 10).
RC-10	Stream Migration activity on 8.8	The page is published for '25 and '26 and states the activity is in 8.7 and later [R15] .	Prove the REST endpoint on 8.8 in the first Release 1 rehearsal (V-K-04).
RC-11	Default customerDeployment	Pega page: generated automatically by the Helm chart when not specified [R16] ; Helm README: defaults to the namespace name [R28] .	Always set it explicitly.

Appendix A. Configuration inventory

Record every environment-specific value here before deployment and keep this appendix under change control. The proposed values follow the naming convention used in this document; replace them only through the change process. Environment codes used below: `dev`, `tst`, `stg`, `prd`.

A.1 Kafka and stream service

Table 68. Kafka inventory values

Parameter	Where set	Proposed value	Recorded value and date
Confluent environment and cluster ID	Confluent Cloud	One cluster for non-production, one for production (OD-05)	
Bootstrap server	stream.bootstrapServer; 8.8 stream settings in Release 1	<bootstrap-host>:9092 from the cluster settings page	
Network connection	Confluent network and Azure private endpoint	Private Link (AD-07)	
Service account	Confluent Cloud	sa-peg-a-<env>, one per environment	
API key ID (not the secret)	Key Vault secret <env>-confluent-jaas	Key ID only in this table	
Topic prefix, Release 1	streamNamePattern	peg-a-<env>-{stream.name}	
Topic prefix, Releases 2 and 3	stream.streamNamePattern	peg-a-<env>2-{stream.name}	
ACL prefixes	Confluent ACLs	Topic and group ACLs for both prefixes while both are in use	
Replication factor	stream.replicationFactor	3	
max.message.bytes	Topic configuration	5000000	
Partition budget	Calculation (Section 7.6)	Measured count under each prefix plus about 100 for Hazelcast removal topics, plus 30 % headroom	

A.2 Search and SRS

Table 69. Search inventory values

Parameter	Where set	Proposed value	Recorded value and date
Search provider and version	Provider contract	A version on the SRS matrix (Section 8.2), for example OpenSearch 2.19 or Elasticsearch 8.18.3	
Search endpoint	srs.srsStorage.domain, port, protocol	<search-host>, 443, https	
SRS user	Search cluster security	srs-<env> with the manage cluster privilege, or settings applied manually	

Parameter	Where set	Proposed value	Recorded value and date
SRS image and tag	srs.srsRuntime.srsImage	search-n-reporting-service-os or search-n-reporting-service, tag from the chart README	
SRS replicas	srs.srsRuntime.replicaCount	3 in production	
customerDeploymentId, Release 1	SRS connection on 8.8	<env>	
customerDeploymentId, Releases 2 and 3	global.customerDeploymentId	<env> - r2	
OAuth token endpoint and JWKS URL	pegasearch.srsAuth.url; srsRuntime.env.OAuthPublicKey URL	From the identity provider (OD-16)	
OAuth client ID	pegasearch.srsAuth.clientId	One client per environment	
Search snapshot repository	Search provider	Provider-managed snapshots, retention per backup policy	

A.3 Pega, AKS and Azure

Table 70. Pega and platform inventory values

Parameter	Where set	Proposed value	Recorded value and date
8.8 exact patch	Source system	From the assessment (Section 4.1)	
Bridge release patch	Installer and Pega images	Latest '24.1 patch, 24.1.4 or later (OD-02)	
Helm chart version	Pipeline	The chart release that supports the target Pega version	
Rules and data schema names	global.jdbc.rulesSchema, dataSchema	Per DBA convention	
Release 3 target schemas	installer.upgrade.targetRulesSchema, targetDataSchema	New rules schema and temporary data schema	
Namespace names	Kubernetes	pega-<env>, srs-<env>, platform-ops	
Key Vault name	Azure	One vault per environment	
Container registry	Azure	One registry; Pega images mirrored and scanned	
Private DNS zones	Azure	Zones for Confluent, the search provider, the database, Key Vault and the registry	
Application Gateway request timeout	Ingress annotation	Agreed with the application team	

Appendix B. Consolidated Helm values skeletons

These skeletons bring together the sections shown earlier. Key names come from the Pega Helm charts README files [R28, R29] and the default values files in the same repository. Values in angle brackets come from Appendix A. Secrets are never written into these files; each `external_secret_name` points to a Kubernetes secret created by the External Secrets Operator (Section 9). Validate each file with `helm template` before use.

B.1 Pega chart, Release 3 (26.1.1)

values-<env>.yaml (pega chart)

```
global:
  provider: "aks"
  deployment:
    name: "pega"
  actions:
    execute: "upgrade-deploy"          # "deploy" after the update completes
  customerDeploymentId: "<env>-r2"
  jdbc:
    url: "<jdbc-url>"
    driverClass: "<jdbc-driver-class>"
    dbType: "<db-type>"
    driverUri: "<driver-uri>"
    external_secret_name: "pega-db-secret" # keys: DB_USERNAME, DB_PASSWORD
    rulesSchema: "<rules-schema>"
    dataSchema: "<data-schema>"
  docker:
    registry:
      url: "<acr-name>.azurecr.io"
    imagePullSecretNames: []           # AKS pulls from ACR with the kubelet identity
    pega:
      image: "<acr-name>.azurecr.io/platform/pega:26.1.1"
  tier:
    - name: "web"
      nodeType: "WebUser"
      service:
        tls:
          enabled: true
          external_secret_names: ["pega-tier-tls"]
      ingress:
        enabled: true
        domain: "<pega-host>"
        annotations:
          appgw.ingress.kubernetes.io/request-timeout: "<seconds>"
      hpa:
        enabled: true
      pdb:
        enabled: true
        minAvailable: 1
    - name: "batch"
      nodeType: "BackgroundProcessing,Search,Batch,RealTime,\
        Custom1,Custom2,Custom3,Custom4,Custom5,BIX"
      hpa:
        enabled: true
      pdb:
        enabled: true
        minAvailable: 1
  hazelcast:
    enabled: false
    clusteringServiceEnabled: false
  stream:
    enabled: true
    bootstrapServer: "<bootstrap-host>:9092"
    securityProtocol: SASL_SSL
    saslMechanism: PLAIN
    streamNamePattern: "pega-<env>2-{{stream.name}}"
    replicationFactor: "3"
    external_secret_name: "pega-stream-secret"
  pegasearch:
    externalSearchService: true
```

```

externalURL: "https://srs-<env>.srs-<env>.svc.cluster.local"
srsAuth:
  enabled: true
  url: "<idp-token-endpoint>"
  clientId: "<srs-client-id>"
  scopes: "pega.search:full"
  authType: "private_key_jwt"
  privateKeyAlgorithm: "RS256"
  external_secret_name: "pega-srs-oauth"
installer:
  image: "<acr-name>.azurecr.io/platform/installer:26.1.1"
  upgrade:
    upgradeType: "zero-downtime"
    targetRulesSchema: "<new-rules-schema>"
    targetDataSchema: "<temp-data-schema>"

```

Note. The tier list above shows only the keys this runbook changes. Copy the remaining tier keys (replicas, resources, livenessProbe, deploymentStrategy) from the default values file of the chart version in use, then apply the sizes from Section 6.

Check the web tier TLS keys (external_secret_names, keystore and certificate options) against the README of the chart version deployed; they changed across chart releases [R28].

B.2 Pega chart, Release 2 differences (bridge release)

Table 71. Release 2 values that differ from Release 3

Key	Release 2 value	Reason
global.actions.execute	upgrade-deploy, with installer.upgrade.upgradeType: in-place	The copied 8.8 database is updated in place in the new environment; the 8.8 source database is not touched [R28].
Pega and installer images	Bridge release tag, for example 24.1.x	OD-02
hazelcast.enabled, clusteringServiceEnabled	false, false	Embedded Hazelcast settings; removal through the DSS (Section 10.6.2) [R8, R28].
stream.streamNamePattern	pega-<env>2-{stream.name}	AD-05
global.customerDeploymentId	<env>-r2	AD-06

B.3 Backingservices chart (SRS)

values-<env>.yaml (backingservices chart)

```

global:
  k8sProvider: "aks"
  imageCredentials:
    registry: "<acr-name>.azurecr.io"
srs:
  enabled: true
  deploymentName: "srs-<env>"
  srsRuntime:
    replicaCount: 3
    # -os image for OpenSearch; search-n-reporting-service for Elasticsearch
    srsImage: "<acr-name>.azurecr.io/platform-services/search-n-reporting-service-os:<tag>"
  env:
    AuthEnabled: true
    OAuthPublicKeyURL: "<idp-jwks-url>"
  ssl:
    enabled: true
    clientAuthentication: "want"
    keystore:
      file: "srs-keystore.p12"
      type: "PKCS12"
    truststore:
      file: "srs-truststore.jks"

```

```
    type: "JKS"
    certsSecret: "srs-runtime-certs"
srsStorage:
  provisionInternalESCluster: false
  domain: "<search-host>"
  port: 443
  protocol: https
  tls:
    enabled: true
  basicAuthentication:
    enabled: true
  authSecret: "srs-search-credentials"
  requireInternetAccess: false
  networkPolicy:
    enabled: true
```

Appendix C. Evidence templates

Each release produces an evidence pack. Store it with the change record. The templates below define what each item contains so that reviewers can compare rehearsal and production results.

C.1 Evidence pack contents

Table 72. Evidence pack items

Item	Content	Produced by	Used for
E-01 Values diff	Diff of Helm values against the previous release, secrets excluded	Platform team	Change approval
E-02 Kafka connectivity	Output of commands K-1 to K-3 from a pod in the Pega namespace	Kafka engineer	V-K-01
E-03 ACL export	Output of the ACL list for the service account	Kafka engineer	V-K-02
E-04 Drain record	GET responses of the Stream Migration activity until COMPLETED, with timestamps	Pega LSA	V-K-04
E-05 Stream status	Screenshot of the Stream landing page: Provider ExternalKafka, Status NORMAL	Pega LSA	V-K-06
E-06 Topic list	Output of K-4 and K-5 for each prefix	Kafka engineer	V-K-07, partition budget
E-07 Search connectivity	Output of S-1 to S-3	Search engineer	V-S-01, V-S-02
E-08 Token check	Decoded token claims (header and payload only)	Security architect	V-S-06
E-09 Index status	Screenshot of the search landing page and output of S-4	Pega LSA	V-S-08 to V-S-10
E-10 Hazelcast checks	hazelcast/disabled and checkRemoteExecutionConnectivity results	Pega LSA	H-6, H-7
E-11 Test report	Smoke and regression results	Test lead	Go/No-Go
E-12 Timings	Start and end time of each step in the release sequence	Release manager	Window planning

C.2 Validation record

Table 73. Validation record template

Field	Entry
Check ID	For example V-K-06
Environment and release	For example stg, Release 2 rehearsal 1
Executed by and date	
Command or screen	
Expected result	Copied from the validation matrix
Actual result	
Result	Pass, Fail, Pass with note
Evidence reference	Evidence pack item number
Deviation and action	Required if the result is not Pass

C.3 Go/No-Go record

Table 74. Go/No-Go record template

Gate	Criteria met	Evidence	Decision owner	Decision and time
Gate 1: readiness			Release manager	
Gate 2: services verified			Platform lead	
Gate 3: update complete			Pega LSA	
Gate 4: business acceptance			Business owner	

Appendix D. Command reference

Run these commands from a tooling pod in the Pega namespace, so they use the same network path, DNS and egress rules as Pega. Use an image mirrored into the customer registry. Read secrets from Key Vault at run time and do not leave them in shell history. Check option names against the installed tool version before use.

D.1 Kafka and Confluent Cloud

Kafka commands

```
# K-1 DNS: the bootstrap host must resolve to a private IP
nslookup <bootstrap-host>

# K-2 TLS: certificate chain and protocol
openssl s_client -connect <bootstrap-host>:9092 -servername <bootstrap-host> </dev/null

# K-3 Authentication and metadata (kcat)
kcat -b <bootstrap-host>:9092 -X security.protocol=SASL_SSL -X sasl.mechanisms=PLAIN \
  -X sasl.username="$API_KEY" -X sasl.password="$API_SECRET" -L

# K-4 Topics under a prefix (Confluent CLI)
confluent kafka topic list --cluster <lkcd-id> | grep "pega-<env>2-"

# K-5 Partition count under a prefix
kcat -b <bootstrap-host>:9092 -X security.protocol=SASL_SSL -X sasl.mechanisms=PLAIN \
  -X sasl.username="$API_KEY" -X sasl.password="$API_SECRET" -L -J \
  | jq --arg p "pega-<env>2-" \
    '[.topics[] | select(.topic | startswith($p)) | .partitions | length] | add'

# K-6 Topic configuration
confluent kafka topic describe <topic> --cluster <lkcd-id>
confluent kafka topic update <topic> --config max.message.bytes=5000000 \
  --cluster <lkcd-id>

# K-7 ACLs for the service account
confluent kafka acl list --service-account <sa-id> --cluster <lkcd-id>
```

D.2 Pega stream migration

Pega documents these requests against a local Pega instance [\[R15\]](#). Send them from inside the Pega pod (for example with `kubectl exec`) or adjust the host to reach one node. Authenticate as the Pega documentation describes for the release in use.

Stream Migration activity [\[R15\]](#)

```
# Start the drain
curl --request POST \
  http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration
# Check status. The drain is complete when queueSize is 0, timeToDrainMS is 0
# and migrationStatus is COMPLETED
curl --request GET \
  http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration
# Restart Kafka queues (only when switching between two external Kafka providers)
curl --request DELETE \
  http://localhost:8080/prweb/PRRestService/CloudRemoteAPI/v1/pzstream/migration
```

D.3 Search and SRS

Search commands

```
# S-1 Version of the search cluster
curl -s -u "$SRS_USER:$SRS_PASSWORD" https://<search-host>/

# S-2 Cluster health
curl -s -u "$SRS_USER:$SRS_PASSWORD" "https://<search-host>/_cluster/health?pretty"
```

```
# S-3 Settings required by SRS
curl -s -u "$SRS_USER:$SRS_PASSWORD" "https://<search-host>/_cluster/settings?pretty"

# S-4 Indexes under a customerDeploymentId prefix
curl -s -u "$SRS_USER:$SRS_PASSWORD" \
  "https://<search-host>/_cat/indices/<deployment-id>?*v"

# S-5 TLS to SRS from the Pega namespace
openssl s_client -connect srs-<env>.srs-<env>.svc.cluster.local:<srs-port> </dev/null
```

D.4 Kubernetes

Kubernetes commands

```
# P-1 Pod status
kubectl get pods -n pega-<env> -o wide

# P-2 Pod logs (stream and search errors)
kubectl logs -n pega-<env> <pod> --since=30m | grep -Ei "stream|kafka|srs|search"

# P-3 Rolling restart of a tier
kubectl rollout restart deployment/<tier-deployment> -n pega-<env>
kubectl rollout status deployment/<tier-deployment> -n pega-<env>

# P-4 External secret synchronization
kubectl get externalsecret -n pega-<env>
kubectl describe externalsecret pega-stream-secret -n pega-<env>

# P-5 Installer job logs
kubectl get jobs -n pega-<env>
kubectl logs -n pega-<env> job/<installer-job> -f
```

D.5 Token inspection

Decode token claims for V-S-06

```
# Decode the JWT payload without verifying it.
# Do not paste production tokens into web tools.
python3 - "$TOKEN" <<'EOF'
import base64, json, sys
payload = sys.argv[1].split(".")[1]
payload += "=" * (-len(payload) % 4)
print(json.dumps(json.loads(base64.urlsafe_b64decode(payload)), indent=2))
EOF
```

D.6 Hazelcast removal checks

These checks run from Dev Studio, not from the command line. Open Records > Integration-Resources > Service Package, open HazelcastDecommission, and in the Methods section run the hazelcast/disabled resource with the GET method and the current requestor context. The expected answer is true. Then run checkRemoteExecutionConnectivity from the same package and confirm that all nodes are reachable [\[R8\]](#).

Appendix E. Glossary

Table 75. Glossary of terms

Term	Meaning
ACL	Access control list. In Kafka, a rule that allows a principal an operation on a resource such as a topic or consumer group.
AKS	Azure Kubernetes Service.
Bridge release	The intermediate Pega release (latest '24.1 patch) used to remove Hazelcast before the update to 26.1.1.
CKU, eCKU	Confluent Unit for Kafka, the capacity unit of Dedicated clusters; elastic CKU for Enterprise clusters.
CodeSet	A Pega rule that groups Java libraries imported into Pega Platform.
customerDeploymentId	The ID that SRS uses to separate the data of each Pega environment. It becomes the index prefix.
Data set (Kafka)	A Pega rule that reads from or writes to a Kafka topic for an application integration. It is separate from the stream service.
DSS	Dynamic System Setting, a Pega configuration record.
External Secrets Operator	A Kubernetes operator that copies secrets from an external store such as Azure Key Vault into Kubernetes secrets.
Hazelcast	The in-memory clustering library used by Pega Platform up to '24.2, removed in '25.
Helm	The package manager for Kubernetes used to deploy the Pega charts.
Jakarta EE	The successor to Java EE. Tomcat 10.1 uses jakarta.* package names instead of javax.*.
JAAS	Java Authentication and Authorization Service. The Kafka client reads its SASL credentials from a JAAS configuration string.
JWKS	JSON Web Key Set, the published public keys that SRS uses to check token signatures.
Private Link	Azure service that exposes a service through a private IP in the customer VNet.
Queue processor	A Pega rule that processes queued items asynchronously through the stream service.
SASL_SSL	Kafka security protocol that combines TLS encryption with SASL authentication.
SRS	Search and Reporting Service, the Pega backing service between Pega Platform and the search cluster.
Stream service	The Pega service that uses Kafka for queue processors, job schedulers and data flows.
streamNamePattern	Helm setting that defines how Pega names its Kafka topics; it carries the environment prefix.
Zero-downtime update	Pega update type that moves rules into a new schema and performs a rolling reboot, so users keep working during the update.

Appendix F. References

Sources were read on the evidence cut-off date in the document control table. Vendor pages change; check each linked page before a release.

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Appendix G. Image sources and attribution

Figures marked "Prepared for this implementation" were drawn for this document from the facts cited in the surrounding text. The other figures are reproduced from the public sources below, unchanged except for scaling, and the combination and numbering of the five zero-downtime schema images.

Table 76. Third-party images

Image	Publisher	Source page
pega_docs_pega-platform-architecture.png	Pega Documentation	[R23]
pega_academy_external_services.png	Pega Academy	[R35]
pega_docs_kafka-use-cases.jpg	Pega Documentation	[R13]
confluent_azure_privatelink.png	Confluent Documentation	[R38]
confluent_azure_dns_records.png	Confluent Documentation	[R38]
pega_academy_three_node.png	Pega Academy	[R36]
pega_academy_srs_cloud_deployment.png	Pega Academy	[R34]
opensearch_cluster.png	OpenSearch Documentation	[R42]
ZDT schema images (five, combined)	Pega Helm charts, GitHub (Apache License 2.0)	[R33]
pega_docs_stream-migration-external-kafka-status.png	Pega Documentation	[R15]

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